Advanced Technologies

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Outline

- Processing Near Memory
- Processing in Memory
- Processing in Sensor
- Processing in the Optical Domain
- Accelerator Level Parallelism

Outline

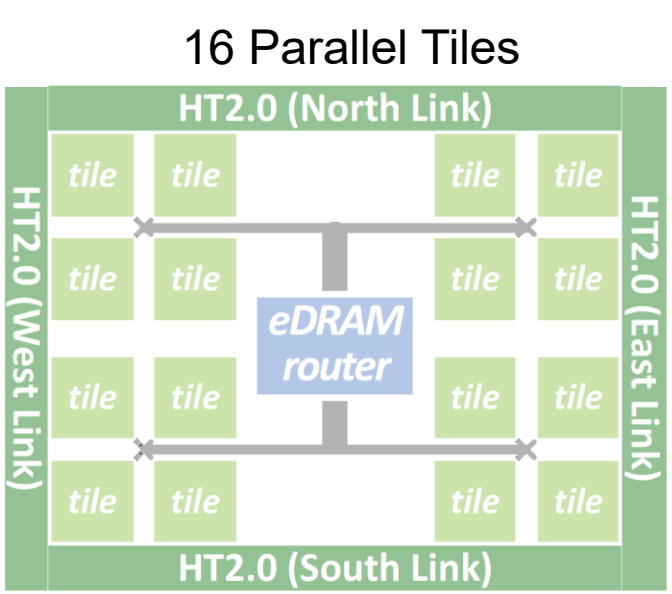
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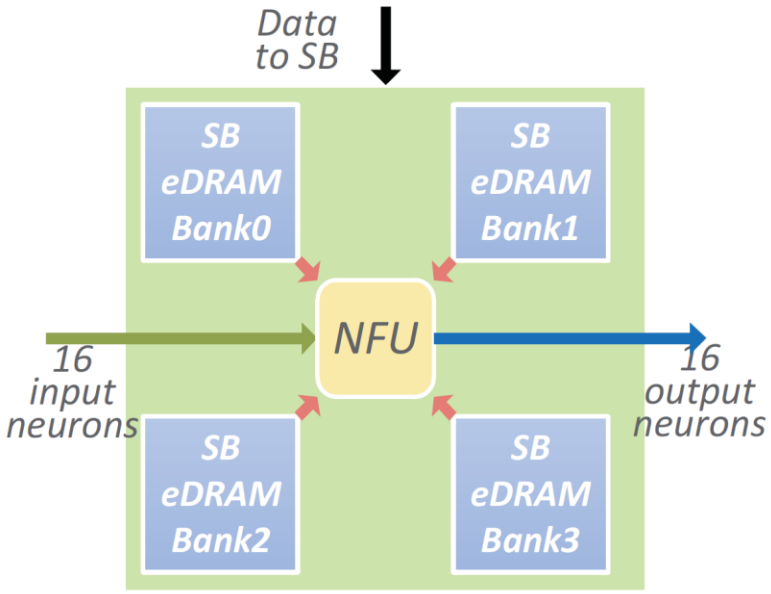
Processing Near Memory

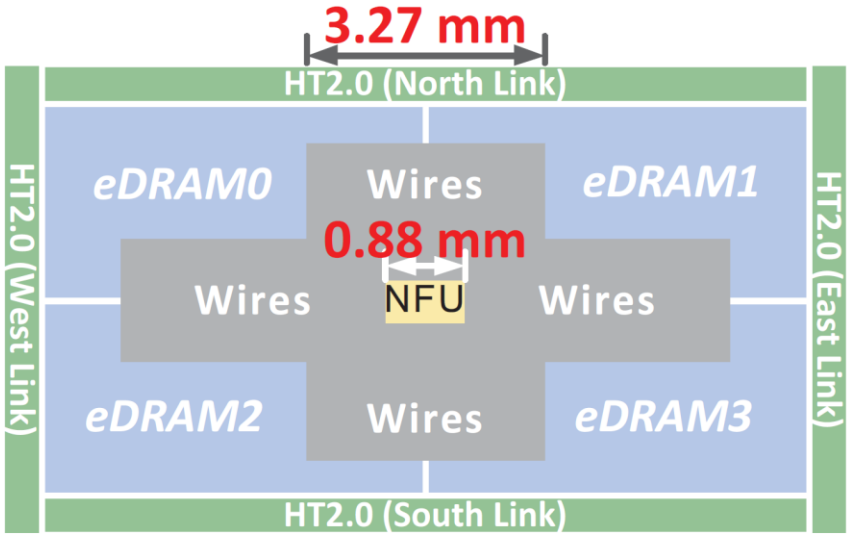
- Bring compute closer to the memory
- Benefits
  - Increase memory bandwidth
  - Reduce energy per access
- Challenges
  - Cost
- Embedded DRAM (eDRAM)
- 3D Stacked Memory (DRAM, SRAM)

eDRAM (DaDianNao)

- Advantages of eDRAM
  - 2.85x higher density than SRAM
  - 321x more energy-efficient than DRAM (DDR3)
- Store weights in eDRAM (36MB)
  - Target fully connected layers since dominated by weights



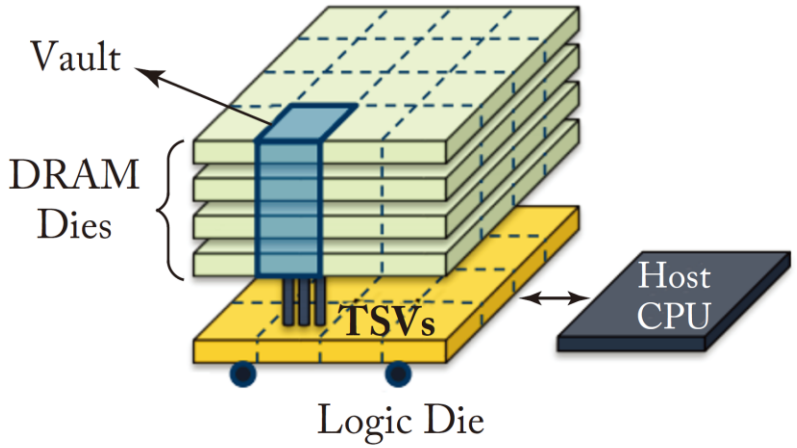


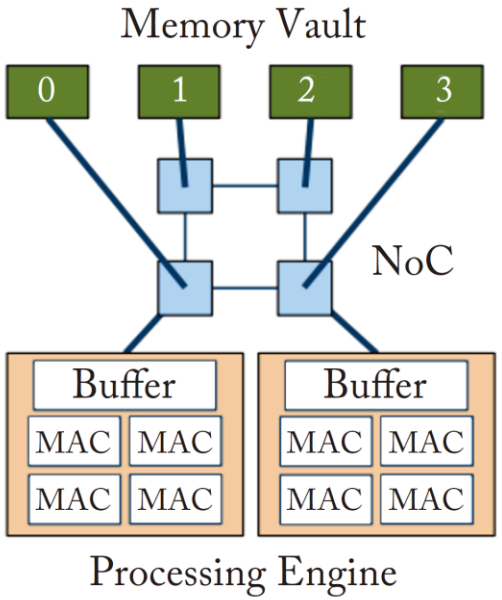


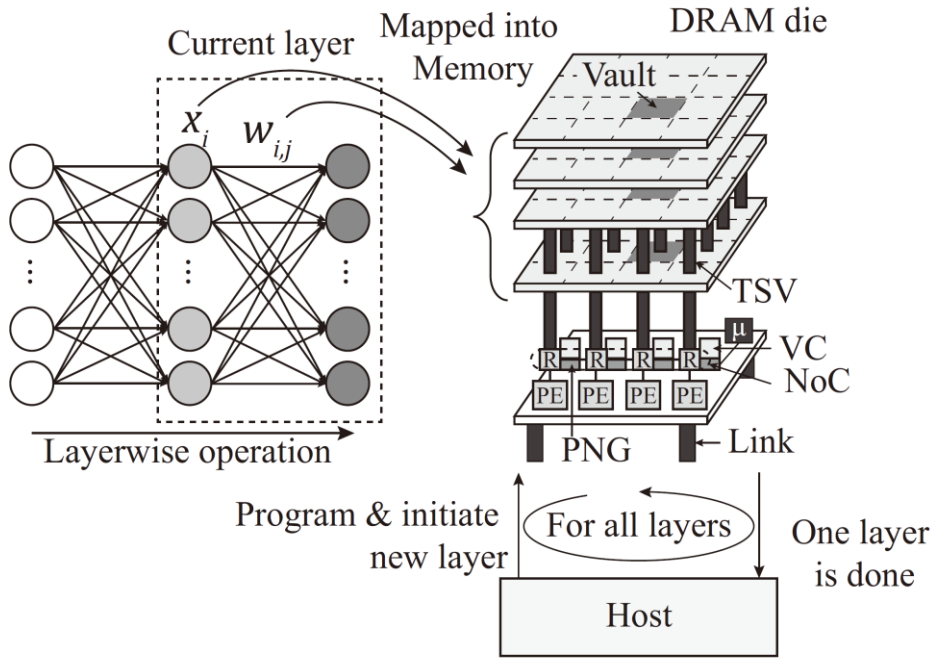
Chen, Y., Luo, T., Liu, S., Zhang, S., He, L., Wang, J., ... & Temam, O. (2014, December). Dadiannao: A machine-learning supercomputer. In *2014 47th Annual IEEE/ACM International Symposium on Microarchitecture* (pp. 609-622). IEEE.

Stacked DRAM (NeuroCube)

- NeuroCube on Hybrid Memory Cube (HMC) Logic Die
  - 6.25x higher BW than DDR3
- HMC (16 ch x 10GB/s) > DDR3 BW (2 ch x 12.8GB/s)
- Computation closer to memory (reduce energy)



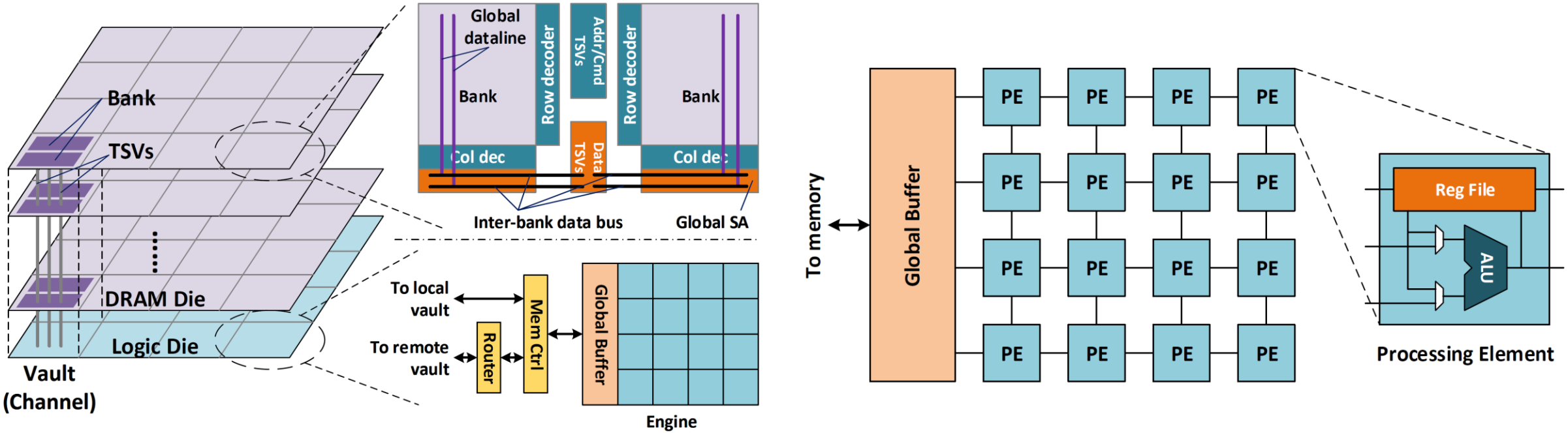




Kim, D., Kung, J., Chai, S., Yalamanchili, S., & Mukhopadhyay, S. (2016). Neurocube: A programmable digital neuromorphic architecture with high-density 3D memory. *ACM SIGARCH Computer Architecture News*, *44*(3), 380-392.

Stacked DRAM (TETRIS)

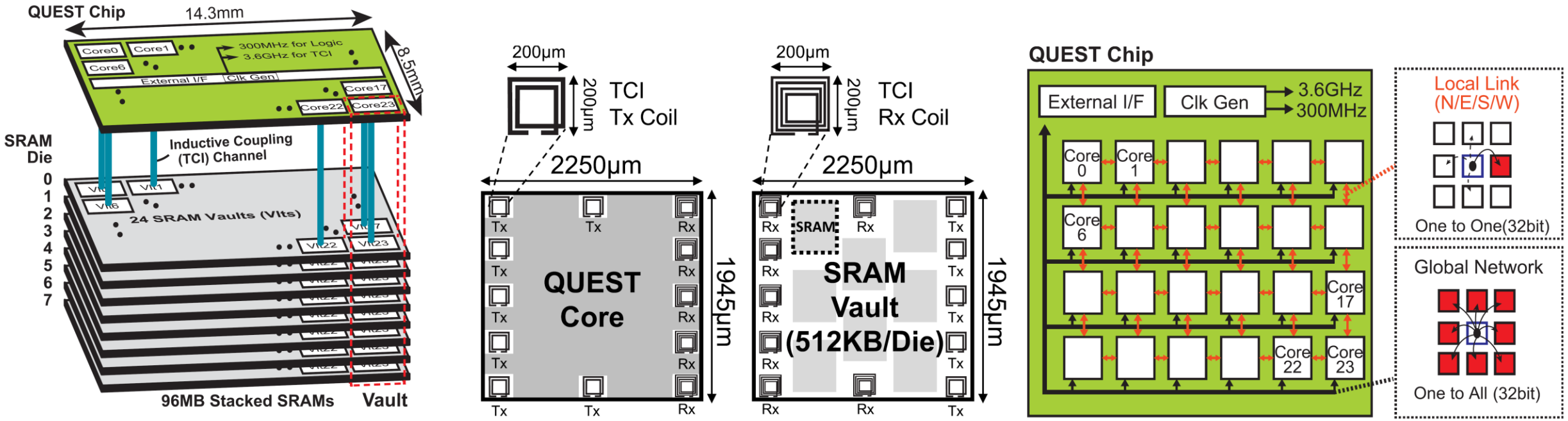
- Explores the use of HMC with the Eyeriss spatial architecture and row stationary dataflow
- Allocates more area to the computation (PE array) than on-chip memory (global buffer) to exploit the low energy and high throughput properties of the HMC
  - 1.5x energy reduction, 4.1x higher throughput vs. 2-D DRAM



Gao, M., Pu, J., Yang, X., Horowitz, M., & Kozyrakis, C. (2017, April). Tetris: Scalable and efficient neural network acceleration with 3d memory. In *Proceedings of the Twenty-Second International Conference on Architectural Support for Programming Languages and Operating Systems* (pp. 751-764).

Stacked SRAM (Quest)

- Explores use of stacked SRAM which has lower latency and power dissipation than DRAM
  - Lower storage density than DRAM but can stack more due to low power density
- Use inductive coupling to connect to SRAM rather than TSV for lower integration cost
  - 9.6Gb/s per channel x 24 channels =28.8 GB/s

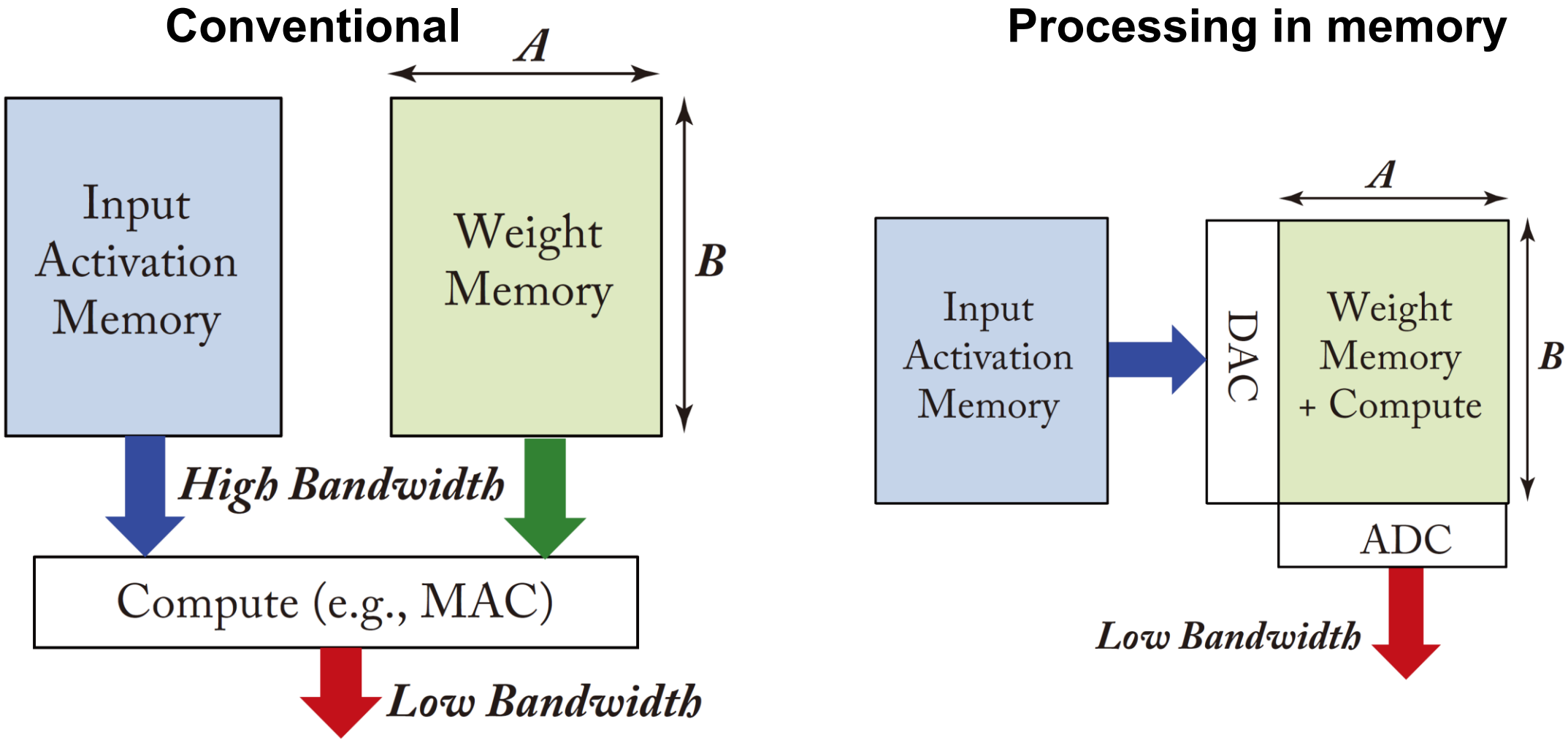


Ueyoshi, K., Ando, K., Hirose, K., Takamaeda-Yamazaki, S., Hamada, M., Kuroda, T., & Motomura, M. (2018). QUEST: Multi-purpose log-quantized DNN inference engine stacked on 96-MB 3-D SRAM using inductive coupling technology in 40-nm CMOS. *IEEE Journal of Solid-State Circuits*, *54*(1), 186-196.

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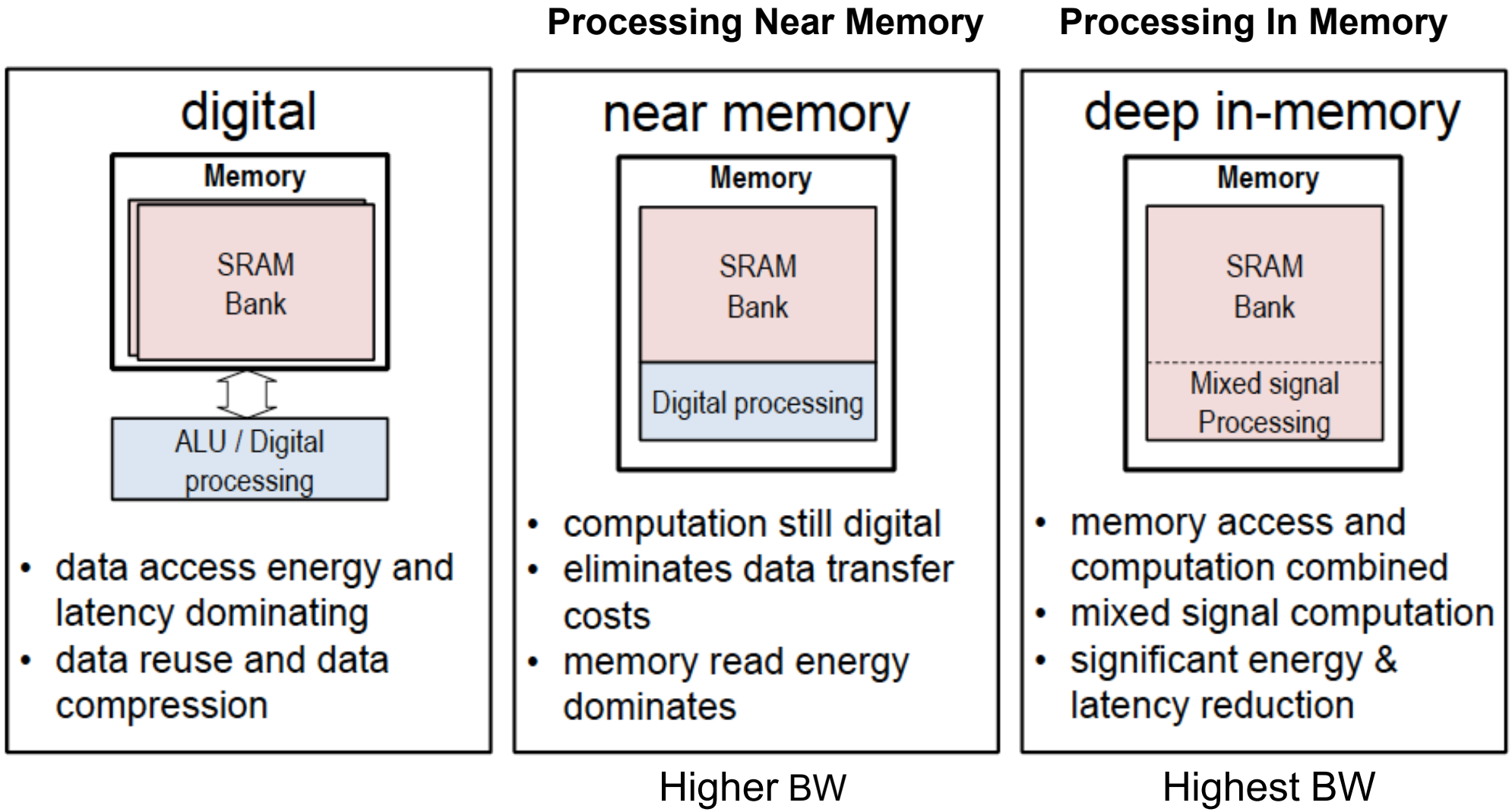
Processing in Memory (PIM)



Processing In Memory

- Embed processing in peripheral or perform compute in analog domain
- Benefits
  - Read multiple weights in parallel (high bandwidth)
  - Perform multiple computations in parallel (high throughput)
  - **Increase density of compute engines** (lower capacitance on input delivery)
  - Reduce number of conversions (sense amp only applied to final accumulated value)
- Challenges
  - **Non-idealities** of analog compute (non-linearity, sensitivity to process and temperature variations)
  - Conversion cost from analog to digital (fewer conversions, but each one is more expensive)
  - Increase size of bit cell and increase size in peripheral (reduce storage density)
  - **Limited precision** that can be stored in each storage element (e.g. bit cell, device); need to use bit serial or multiple elements if want to support higher precision; also may need to use two arrays to enable differential signal to represent negative values
- Memory Technology
  - SRAM, DRAM, NVM (Flash, memristors)

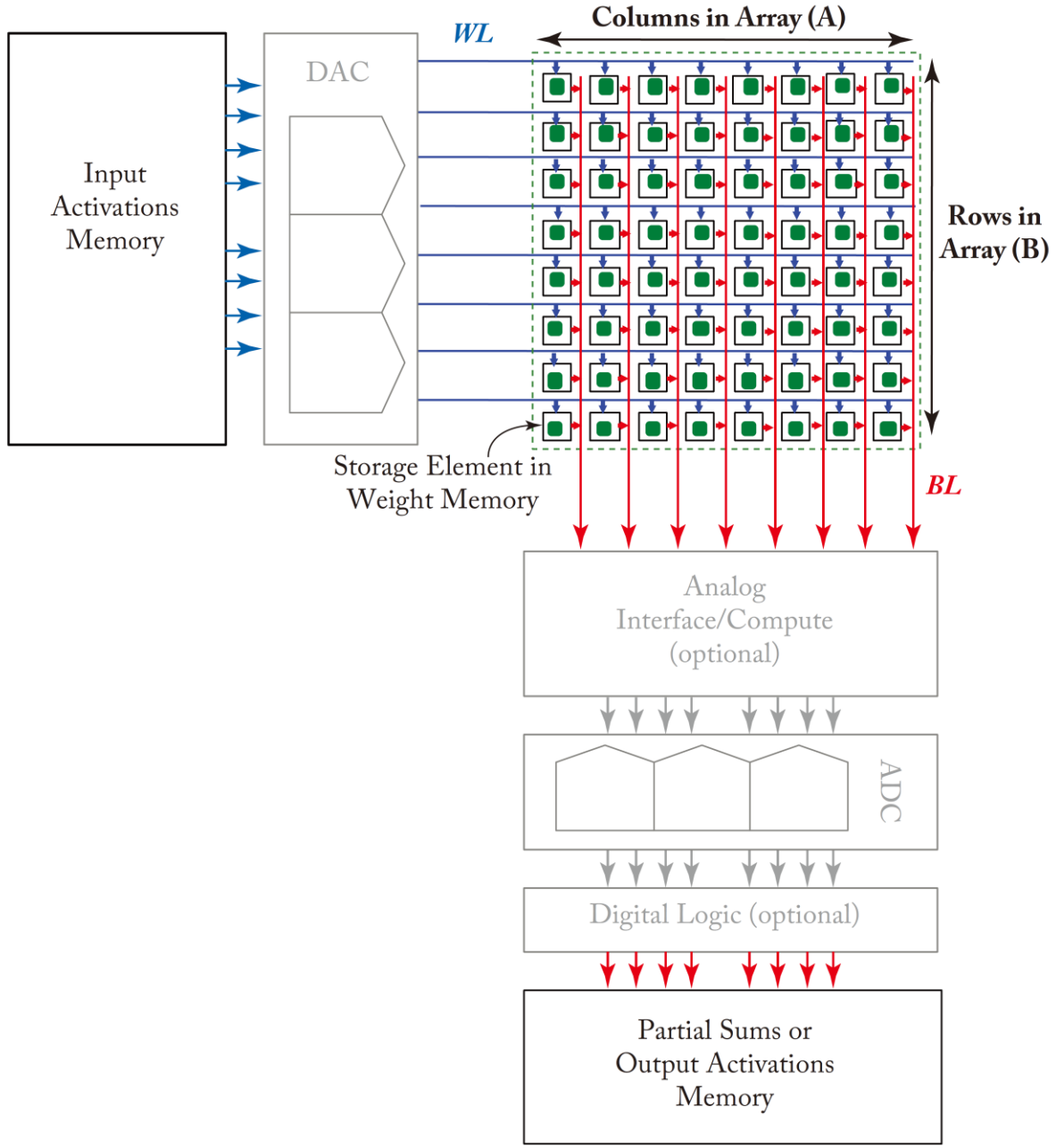
Processing Near vs. In Memory



Gonugondla, S. K., Kang, M., & Shanbhag, N. R. (2018). A variation-tolerant in-memory machine learning classifier via on-chip training. *IEEE Journal of Solid-State Circuits*, *53*(11), 3163-3173.

Dataflow for PIM

- Weight stationary dataflow
- Overhead in conversion from analog to digital and digital back to analog



FC Layer for PIM

```
i = Array(CHW)          # Input activations
f = Array(M, CHW)       # Filter weights
o = Array(M)            # Output partial sums

parallel–for m in [0, M):
    parallel–for chw in [0, CHW):
        o[m] += i[chw] * f[m, chw]
```

1-D Weight-Stationary Convolution Dataflow for PIM

```
i = Array(C, W)          # Input activations
f = Array(M, C, S)       # Filter weights
o = Array(M, Q)          # Output partial sums

parallel-for m in [0, M):
    parallel-for s in [0, S):
        parallel-for c in [0, C]:
            for q in [0, Q):
                w = q + s
                o[m, q] += i[c, w] * f[m, c, s]
```

Analog Computation

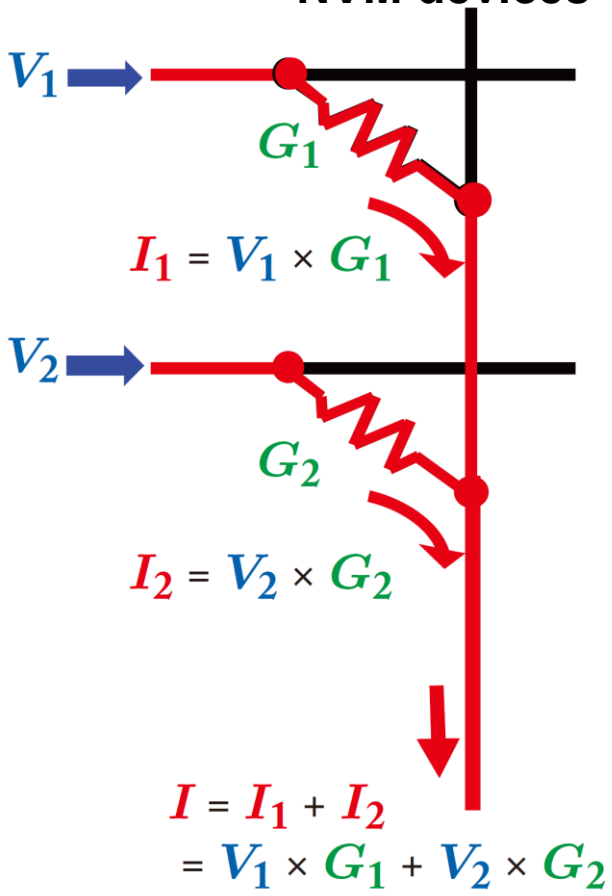
- Conductance = Weight
- Voltage = Input
- Current = Voltage × Conductance
- Sum currents for addition

$$Output = \sum Weight \times Inpt$$

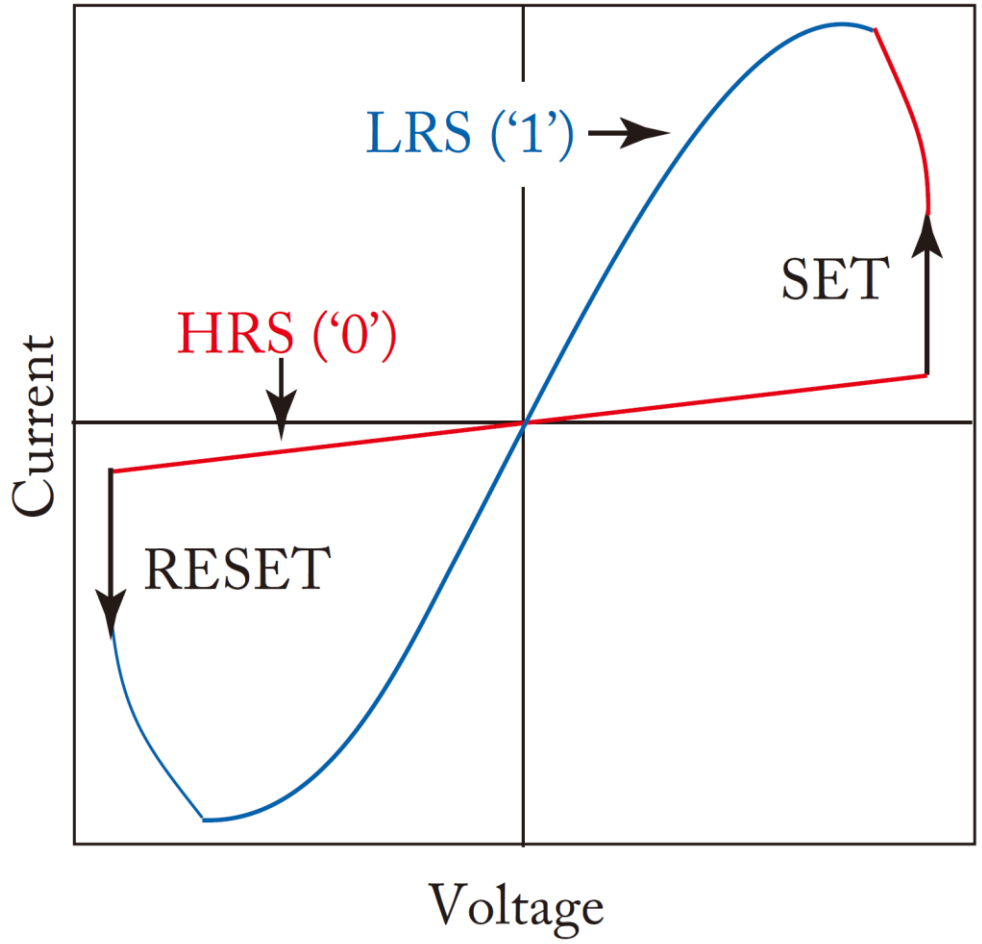
$$Input = V1, V2, \dots (Voltage)$$
$$Weights = G1, G2, \dots (conductance)$$
$$Output = Current\ summation$$

MAC with Resistive Non-Volatile Memories (NVM)

MAC operation using resistive NVM devices

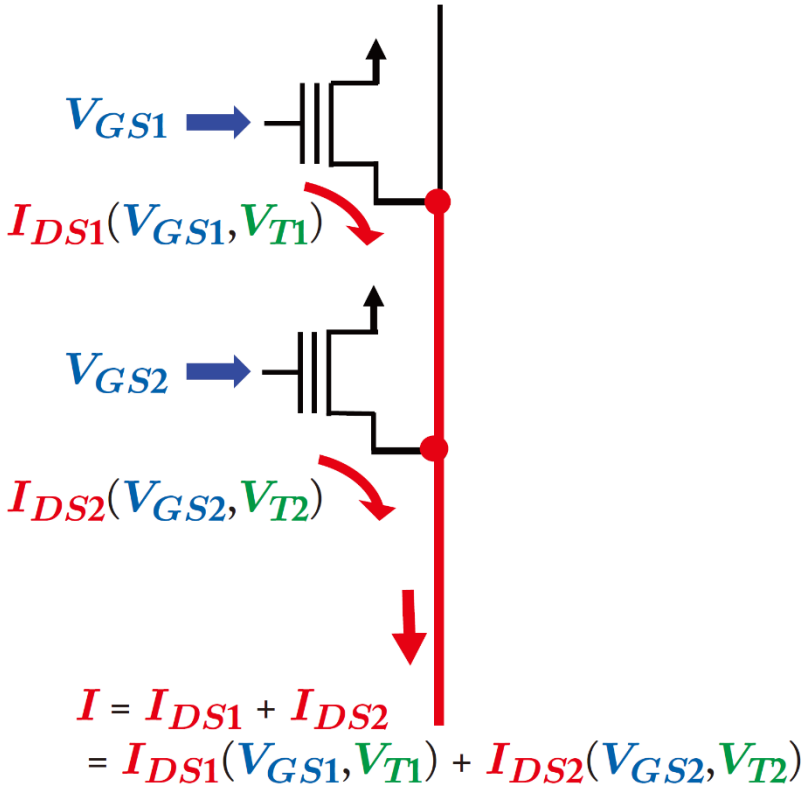


Example I-V curve of a resistive NVM device

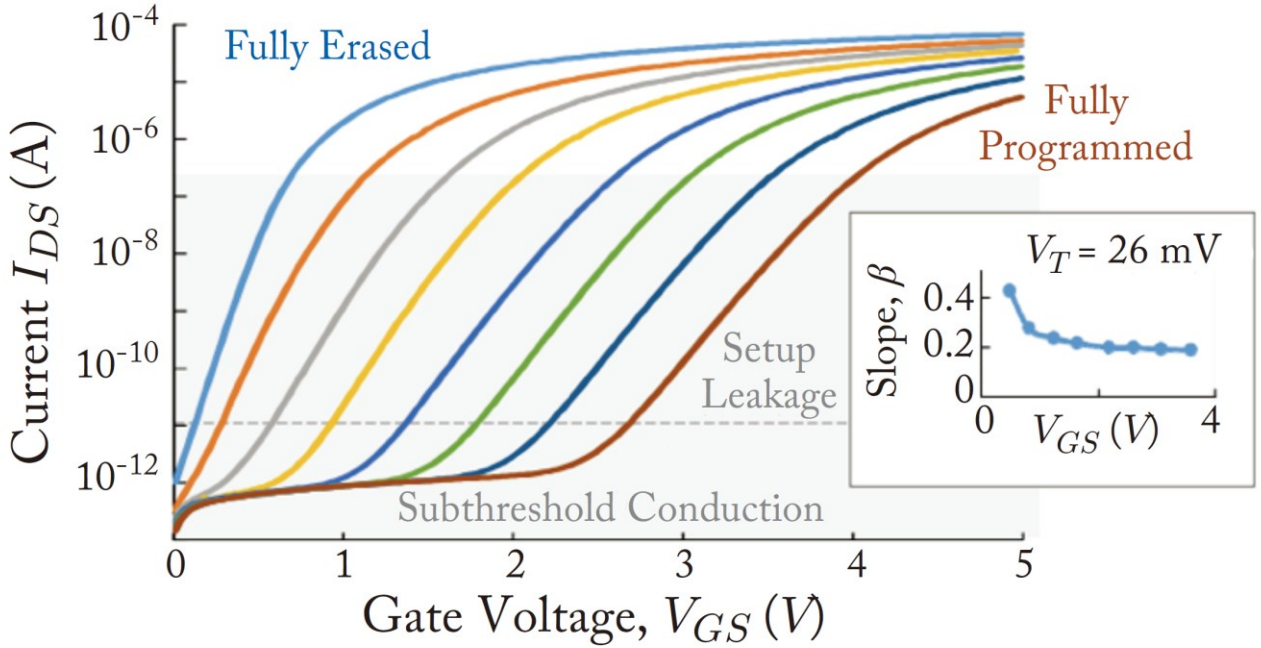


MAC with Floating Gate Non-Volatile Memories (NVM)

MAC operation using floating gate NVM devices

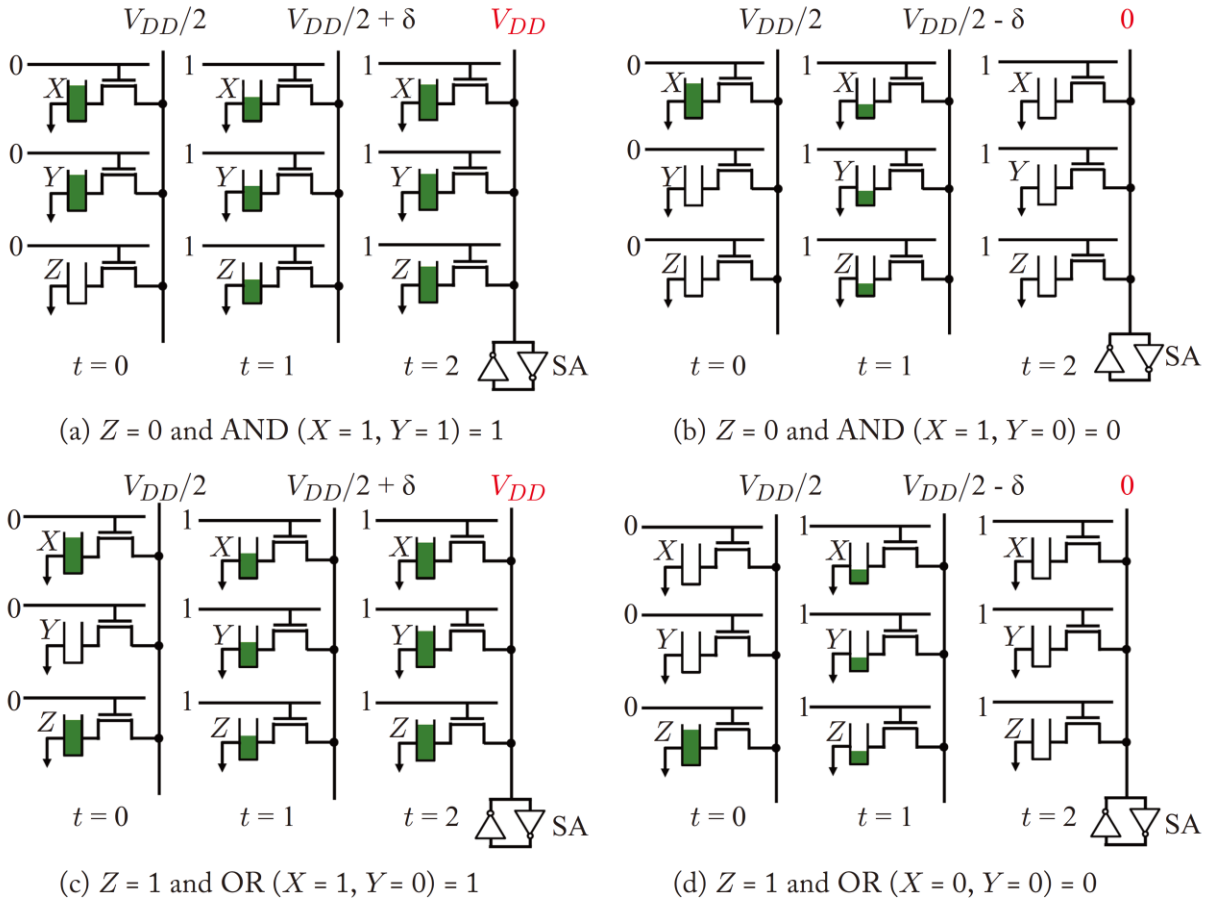


I-V curve of floating gate device



Compute in DRAM Based on Charge Sharing

- Z controls whether an AND or OR is performed on input X and Y



(a) $Z = 0$ and AND $(X = 1, Y = 1) = 1$

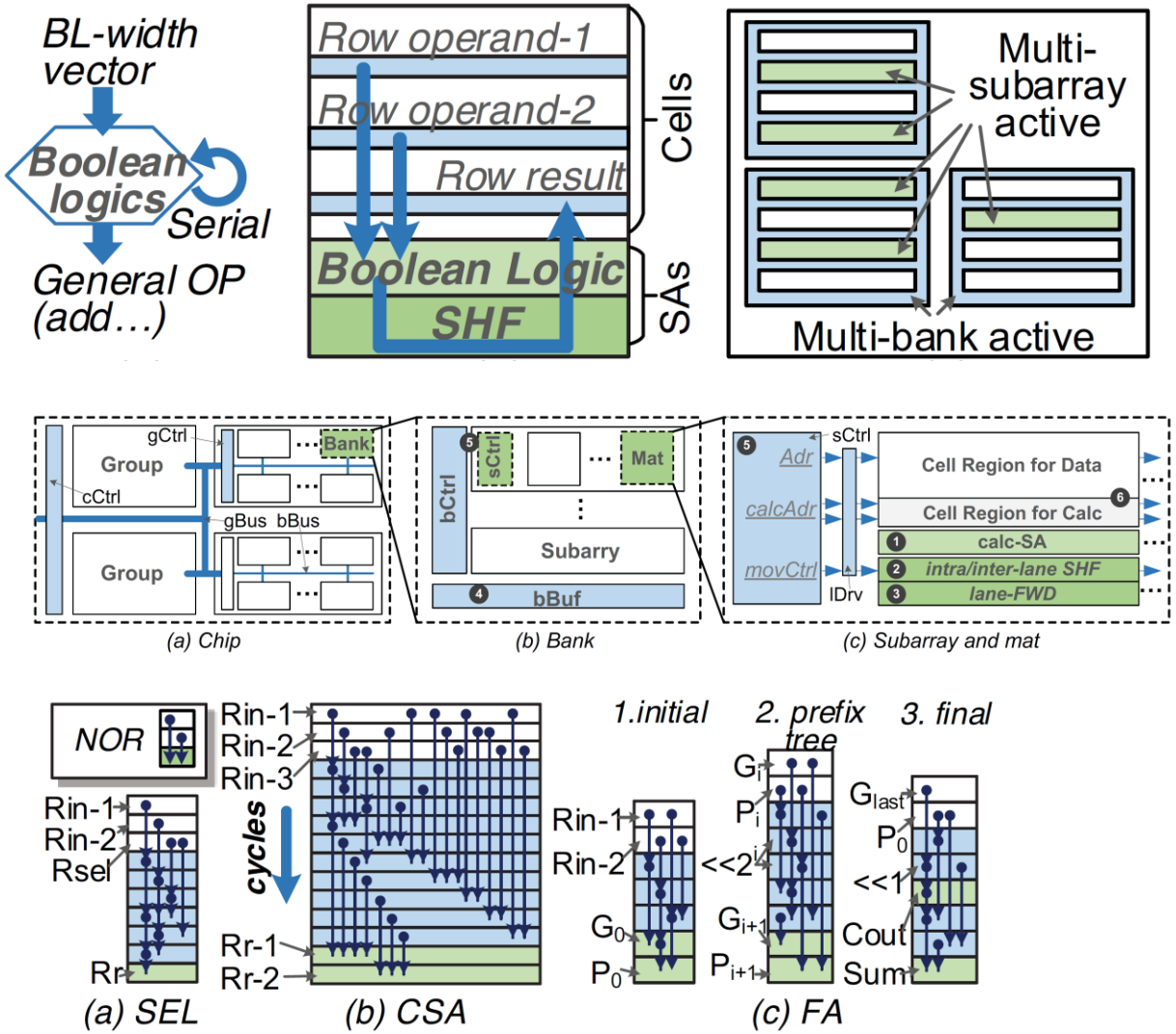
(b) $Z = 0$ and AND $(X = 1, Y = 0) = 0$

(c) $Z = 1$ and OR $(X = 1, Y = 0) = 1$

(d) $Z = 1$ and OR $(X = 0, Y = 0) = 0$

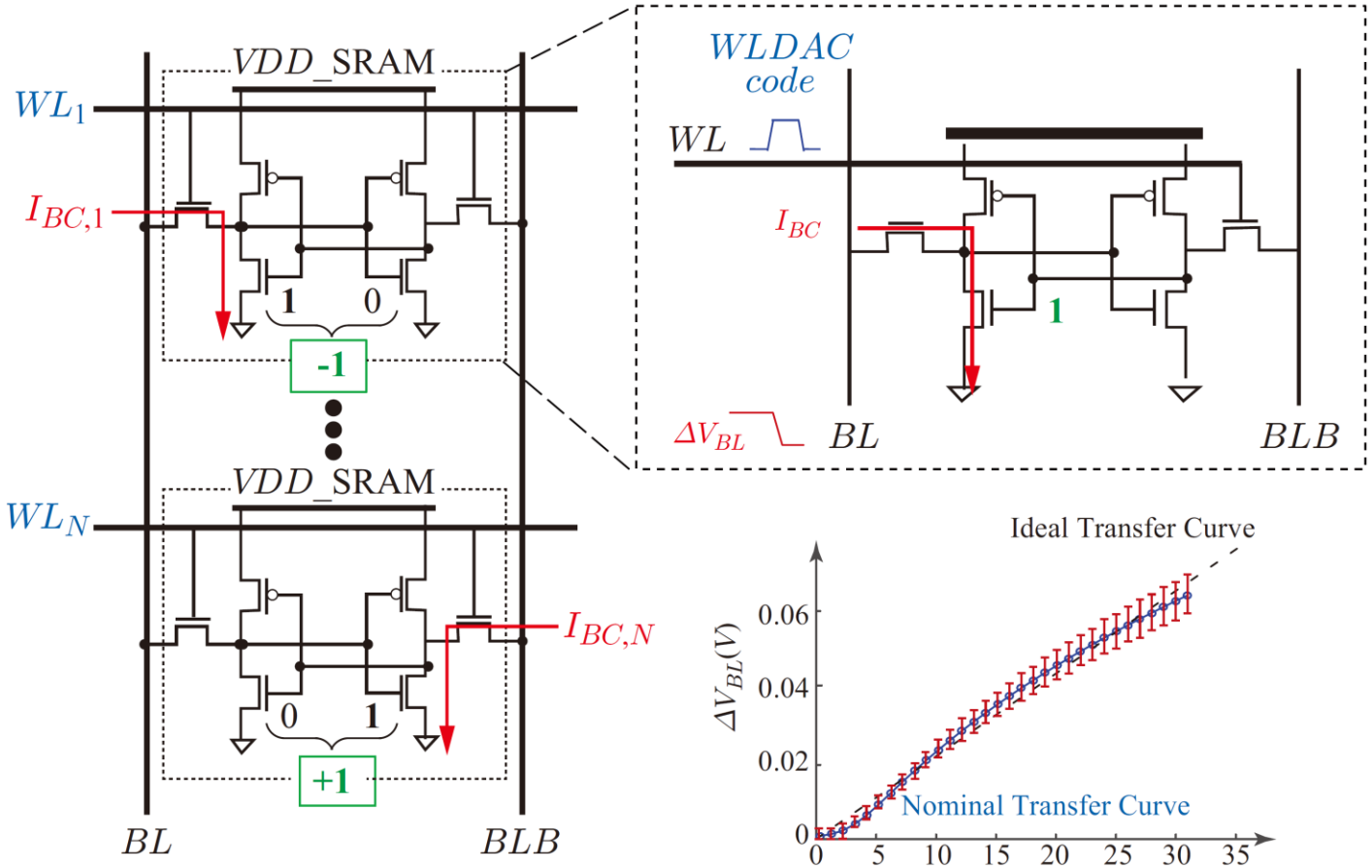
DRAM (DRISA)

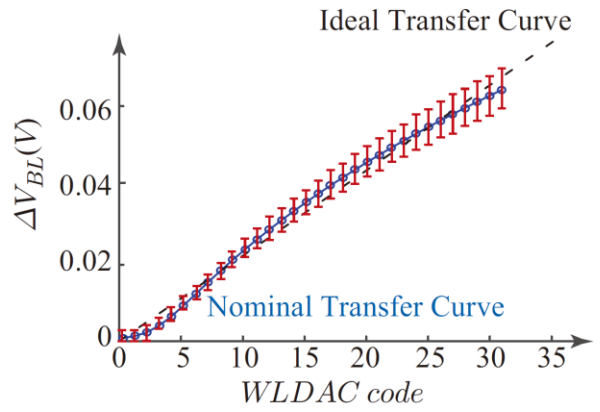
- Build multiplier and adders from NOR logic in DRAM
  - Add shift logic to output of DRAM before write back
- Requires multiple cycles
  - Number of cycles depends on bit width
  - Tradeoff parallelism for cycles
- Parallelism dictated by array width and number of arrays
- Demonstrate on BNN: 1b weights, 8 bit activation for AlexNet, VGG-16, VGG-19, ResNet

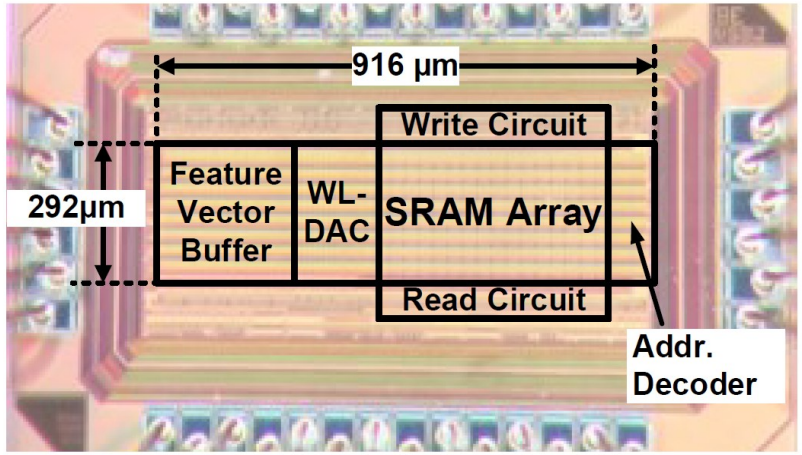


Li, S., Niu, D., Malladi, K. T., Zheng, H., Brennan, B., & Xie, Y. (2017, October). Drisa: A dram-based reconfigurable in-situ accelerator. In *2017 50th Annual IEEE/ACM International Symposium on Microarchitecture (MICRO)* (pp. 288-301). IEEE.

Compute Multiplication in 6T SRAM Array

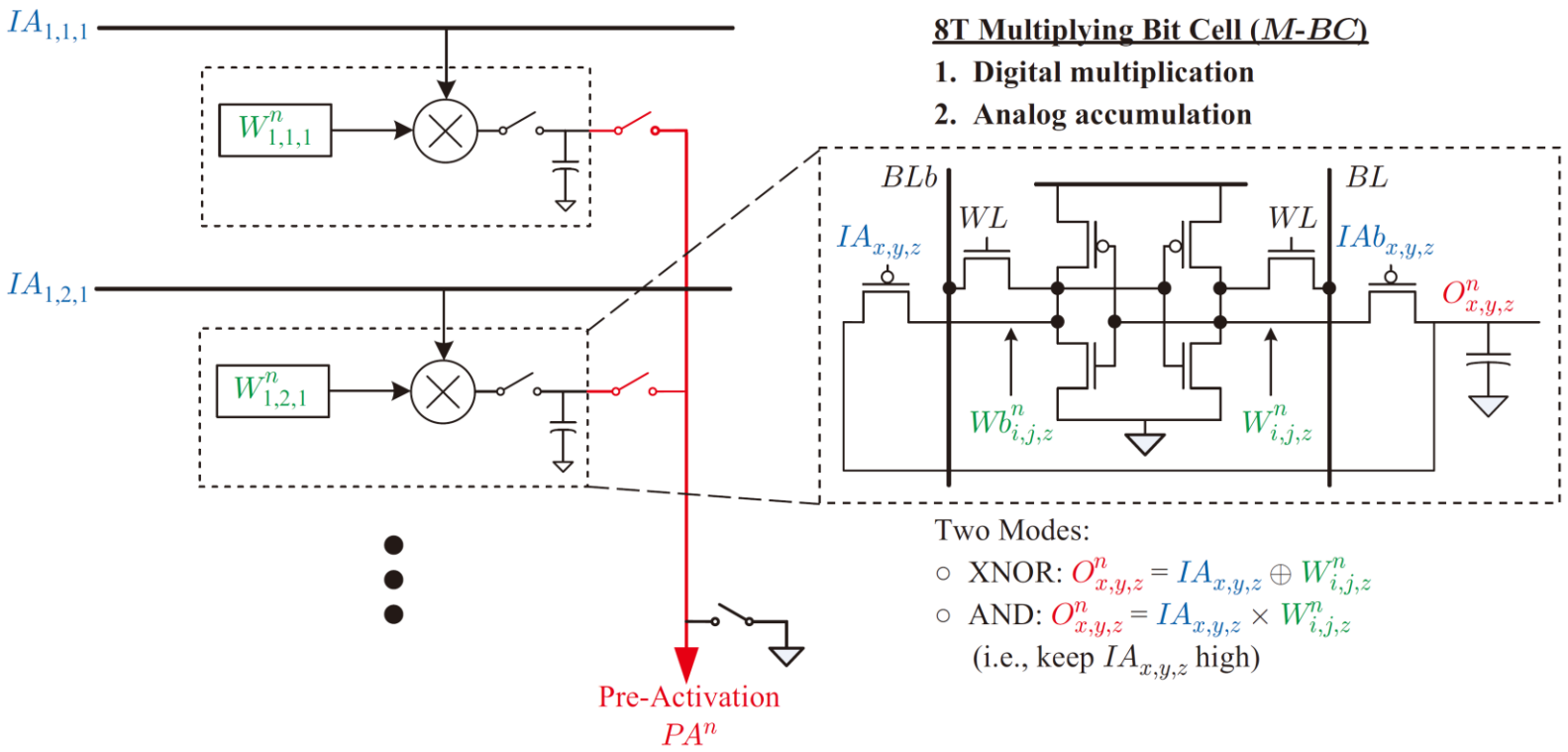


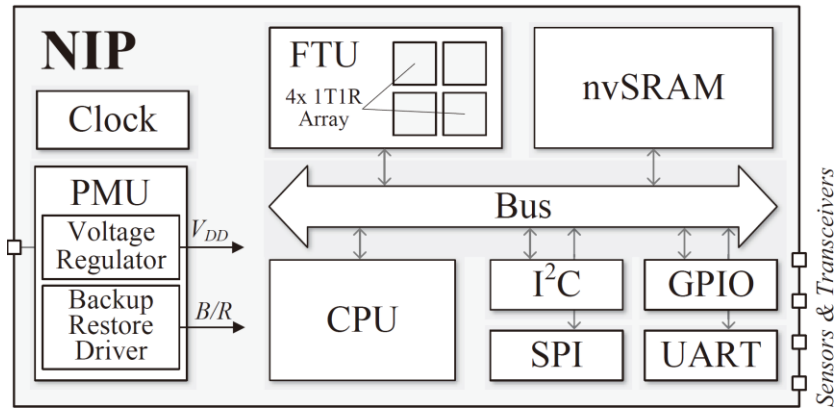




Zhang, J., Wang, Z., & Verma, N. (2016, June). A machine-learning classifier implemented in a standard 6T SRAM array. In *2016 ieee symposium on vlsi circuits (vlsi-circuits)* (pp. 1-2). IEEE.

Compute Multiplication in 8T SRAM Array

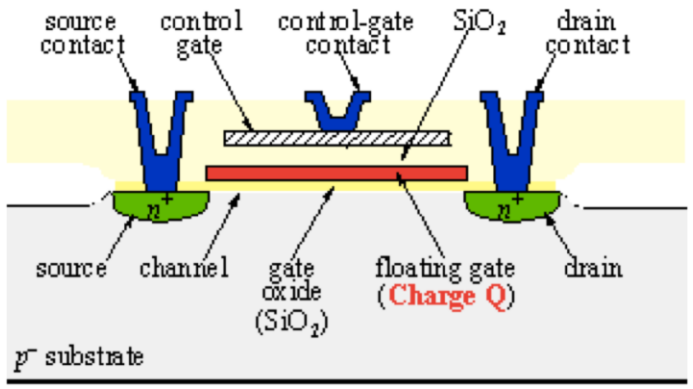




Su, F., Chen, W. H., Xia, L., Lo, C. P., Tang, T., Wang, Z., ... & Liu, Y. (2017, June). A 462GOPs/J RRAM-based nonvolatile intelligent processor for energy harvesting IoE system featuring nonvolatile logics and processing-in-memory. In *2017 Symposium on VLSI Technology* (pp. T260-T261). IEEE.

Analog Compute with Flash Memory

- Floating Gate: Program by changing threshold voltage (shift I-V curve)



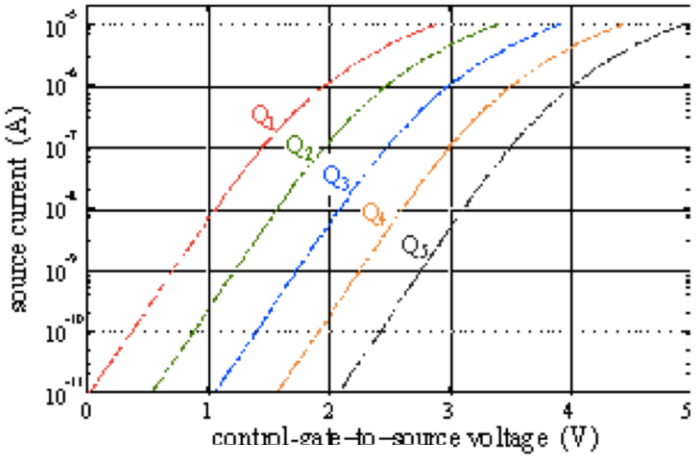


Image source: Diorio, ISFIIE 1999

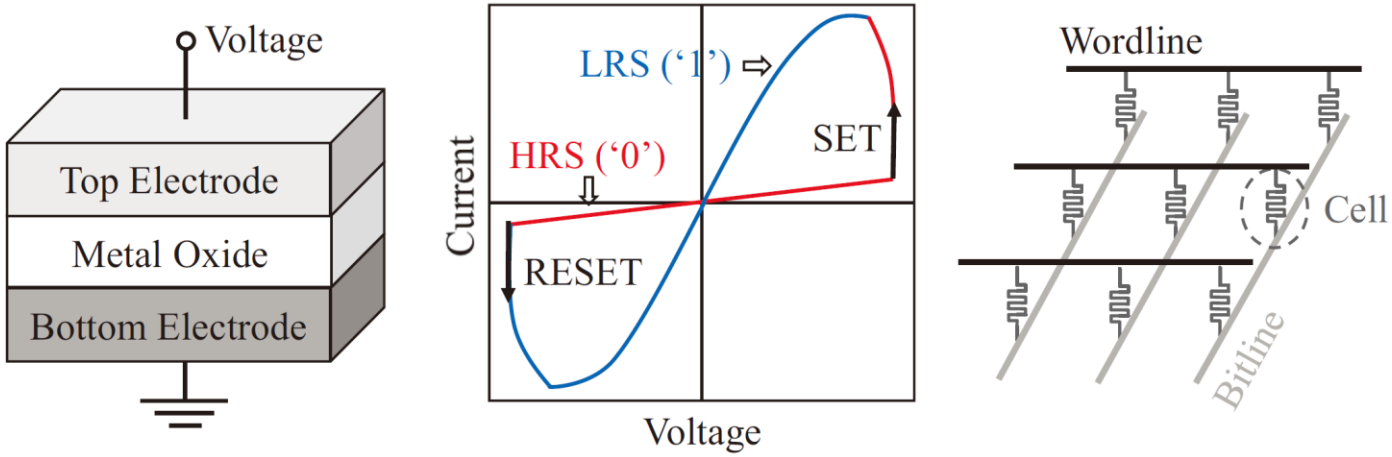
	Enterprise With DRAM	Enterprise No-DRAM	Edge With DRAM	Edge No-DRAM	Mythic NVM
SRAM	<50 MB	100+ MB	< 5 MB	< 5 MB	< 5 MB
DRAM	8+ GB	-	4-8 GB	-	-
Power	70+ W	70+ W	3-5 W	1-3 W	1-5 W
Sparsity	Light	Light	Moderate	Heavy	None
Precision	32f / 16f / 8i	32f / 16f / 8i	8i	1-8i	1-8i
Accuracy	Great	Great	Moderate	Poor	Great
Performance	High	High	Very Low	Very Low	High
Efficiency	25 pJ/MAC	2 pJ/MAC	10 pJ/MAC	5 pJ/MAC	0.5 pJ/MAC

Source: Mythic, Hot Chips 2018

Memristor Computation

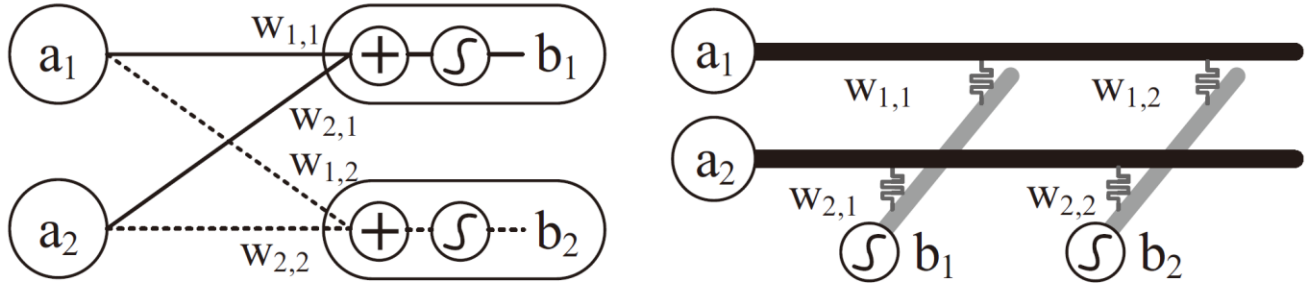
- Use memristors as programmable weights (resistance)
- Advantages
  - High Density (< 10nm x 10nm size*)
    - ~30x smaller than SRAM**
    - 1.5x smaller than DRAM**
- Non-Volatile
  - Operates at low voltage
  - Computation within memory (in situ)
  - Reduce data movement

Memristor



(a) (b) (c)

Figure 1. (a) Conceptual view of a ReRAM cell; (b) I-V curve of bipolar switching; (c) schematic view of a crossbar architecture.



(a) (b)

Figure 2. (a) An ANN with one input/output layer; (b) using a ReRAM crossbar array for neural computation.

Chi, P., Li, S., Xu, C., Zhang, T., Zhao, J., Liu, Y., ... & Xie, Y. (2016). Prime: A novel processing-in-memory architecture for neural network computation in reram-based main memory. *ACM SIGARCH Computer Architecture News*, *44*(3), 27-39.

Challenges with Memristors

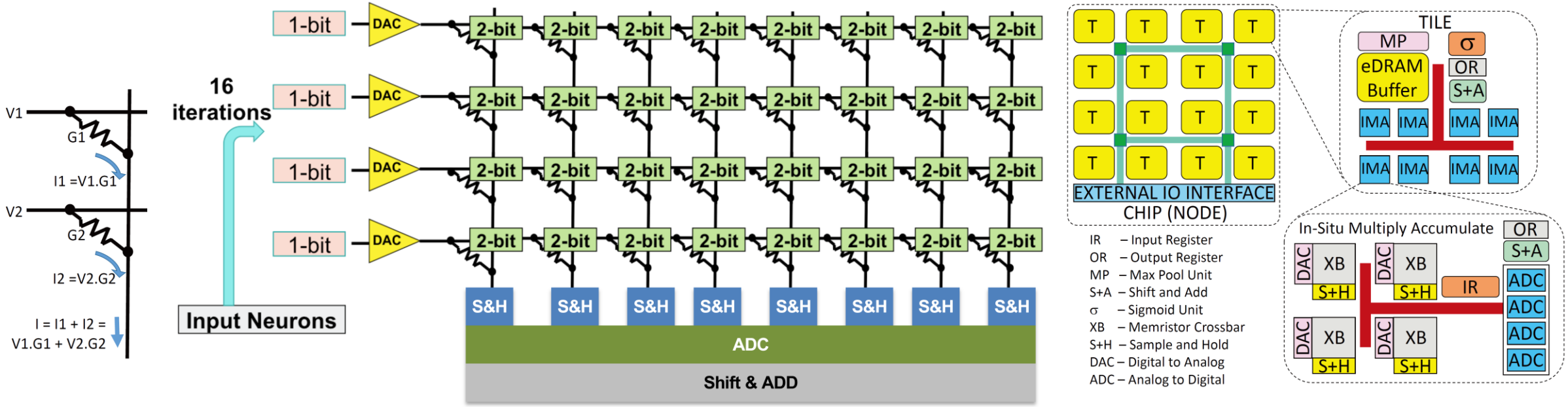
- Limited Precision
- A/D and D/A Conversion
- Array Size and Routing
  - Wire dominates energy for array size of 1k × 1k
  - IR drop along wire can degrade read accuracy
- Write/programming energy
  - Multiple pulses can be costly
- Variations & Yield
  - Device-to-device, cycle-to-cycle
  - Non-linear conductance across range

Eryilmaz, S. B., Joshi, S., Neftci, E., Wan, W., Cauwenberghs, G., & Wong, H. S. P. (2016, March). Neuromorphic architectures with electronic synapses. In *2016 17th International Symposium on Quality Electronic Design (ISQED)* (pp. 118-123). IEEE.

ISAAC

- eDRAM using memristors
- 16-bit dot-product operation
  - 8 x 2-bits per memristors
  - 1-bit per cycle computation
  - Trade off area and cycles to address low precision

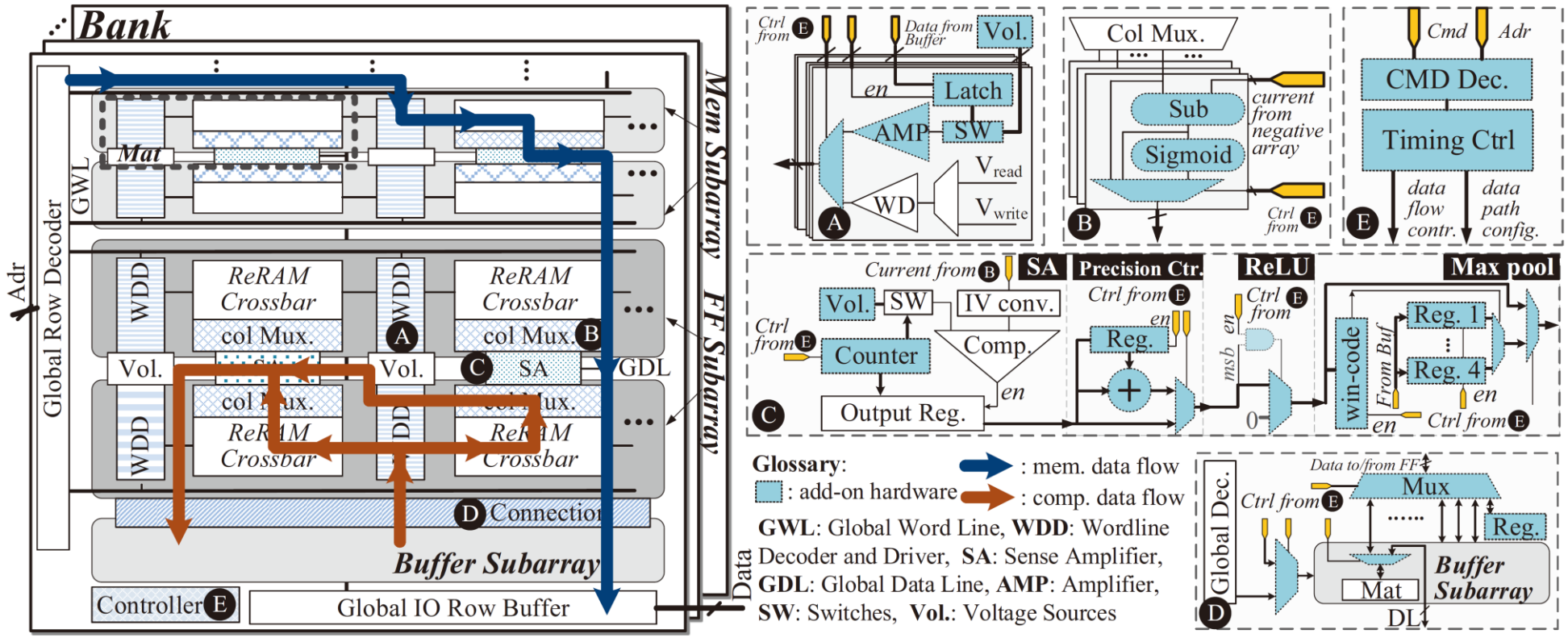
Eight 128x128 arrays per IMA
12 IMAs per Tile
14x12 Tiles in ISAAC



Shafiee, A., Nag, A., Muralimanohar, N., Balasubramonian, R., Strachan, J. P., Hu, M., ... & Srikumar, V. (2016). ISAAC: A convolutional neural network accelerator with in-situ analog arithmetic in crossbars. *ACM SIGARCH Computer Architecture News*, *44*(3), 14-26.

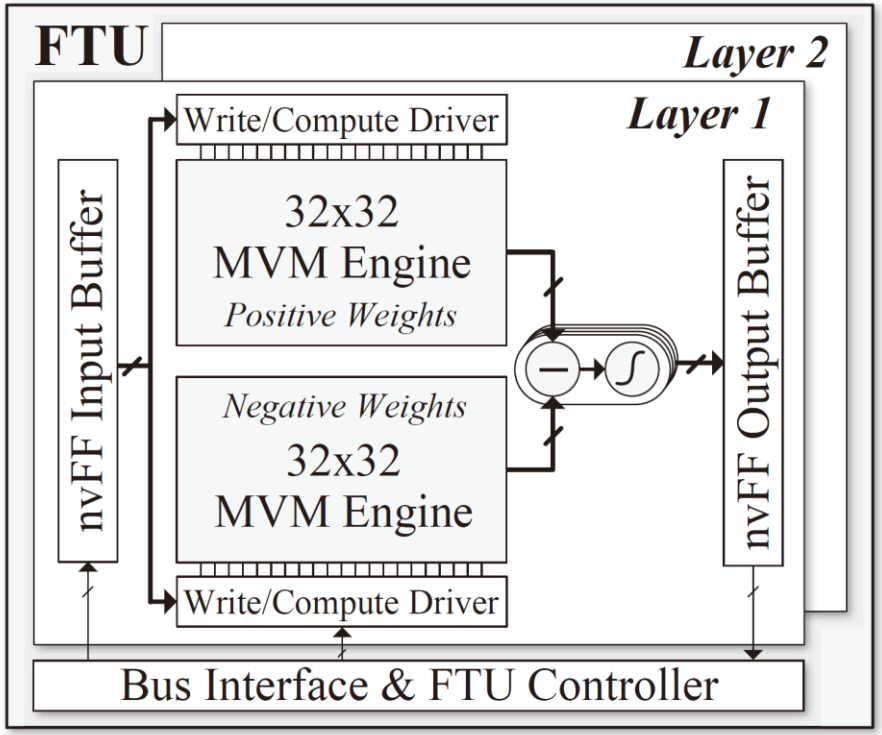
PRIME

- Bit precision for each 256x256 ReRAM array
  - 3-bit input, 4-bit weight (2x for 6-bit input and 8-bit weight)
  - Dynamic fixed point (6-bit output)
- Reconfigurable to be main memory or accelerator
  - 4-bit MLC computation; 1-bit SLC for storage



Fabricated Arrays (FTU)

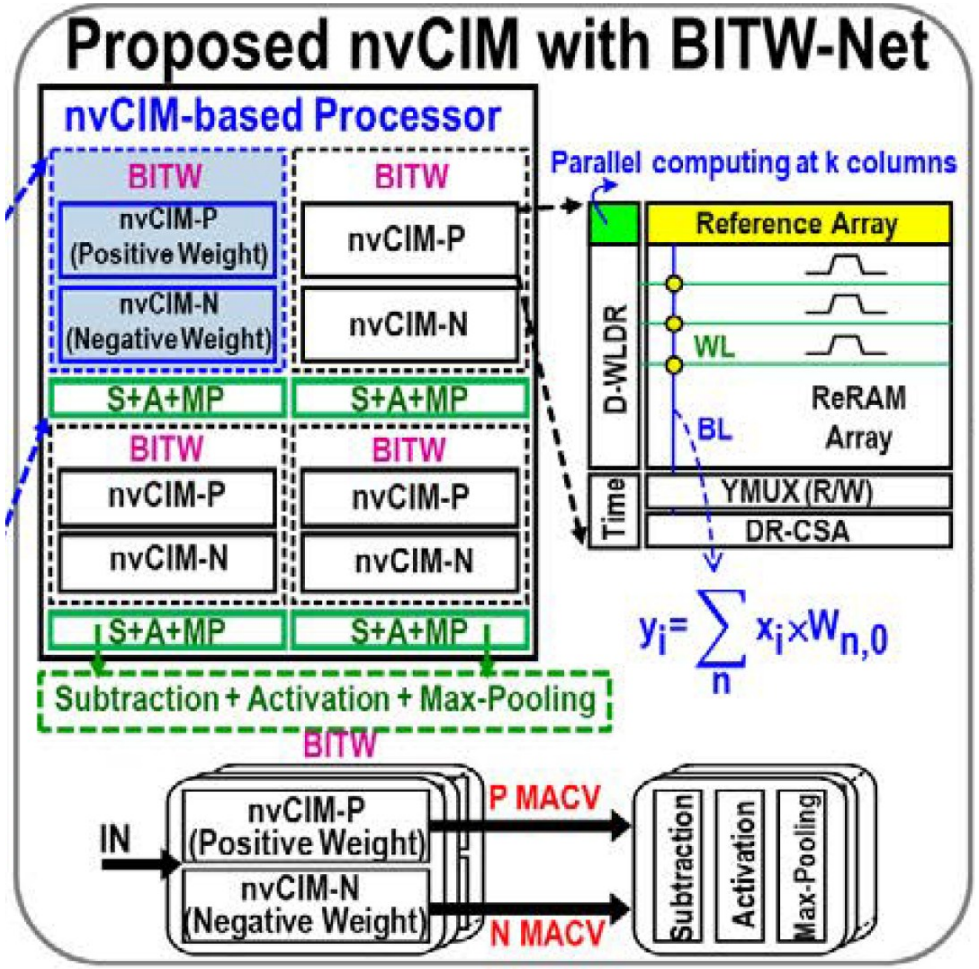
- Ternary weights and binary activations
- Use two arrays for ternary weights {-1, 0, +1}
- Array size: 32x32 (2 arrays)
  - Accumulate 32 values into a 3-bit output
- Demo Task
  - Input: 5x5 image (25 values)
  - Classify with 2 FC (25x32x4)
  - Output: 4 directions



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Fabricated Arrays (nvCIM)

- Ternary weights and binary activations
- Use two arrays for ternary weights {-1, 0, +1}
- Array size: 512 x 256 (8 arrays)
  - Accumulate 512 values into a 3-bit output
  - Max 32 rows activated at a time due to sensing margin and yield
- Demo Task
  - Input: 5x5 image (MNIST) – 96.2% accuracy
  - Classify with 2 FC (25x25x3) ??
  - Output: 9 digits



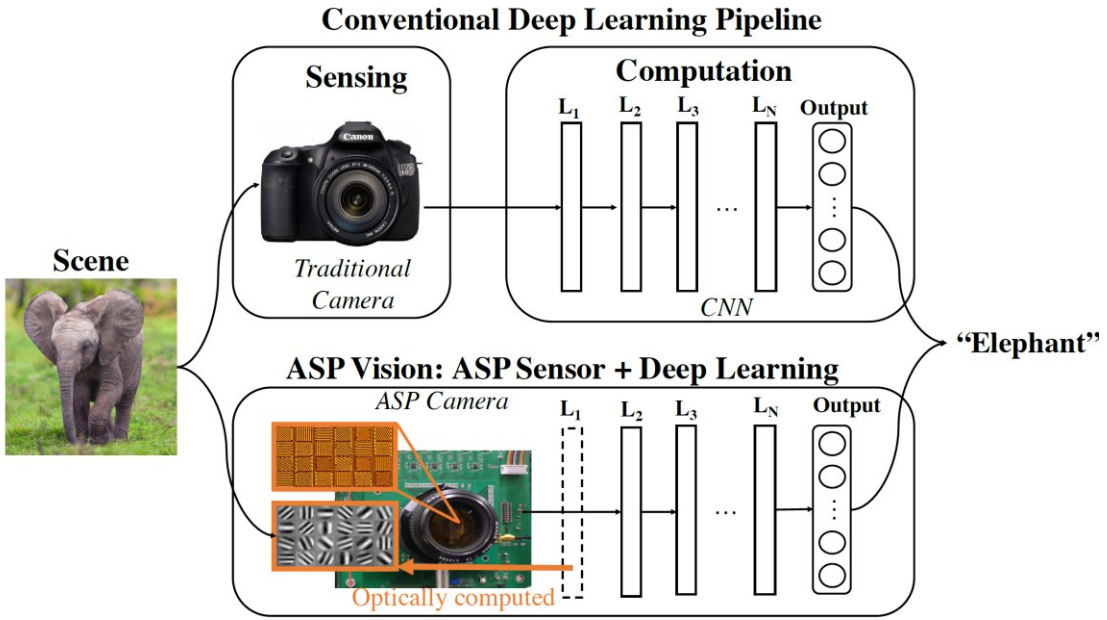
Chen, W. H., Li, K. X., Lin, W. Y., Hsu, K. H., Li, P. Y., Yang, C. H., ... & Chang, M. F. (2018, February). A 65nm 1Mb nonvolatile computing-in-memory ReRAM macro with sub-16ns multiply-and-accumulate for binary DNN AI edge processors. In *2018 IEEE International Solid-State Circuits Conference-(ISSCC)* (pp. 494-496). IEEE.

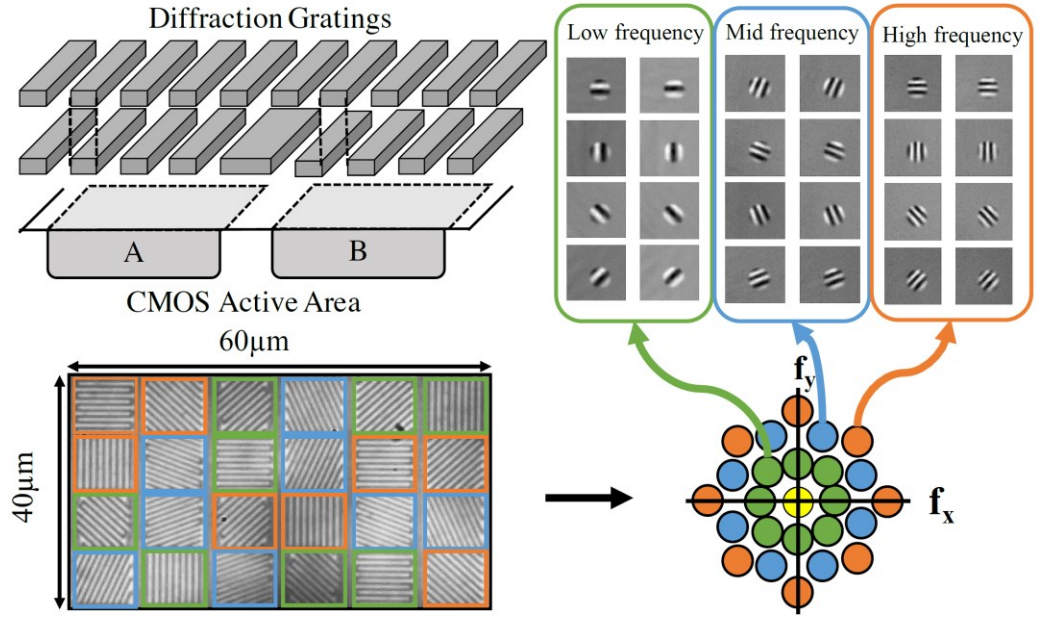
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ASP Vision

- Optically Computing the First Layer of Convolutional Neural Networks using Angle Sensitive Pixels

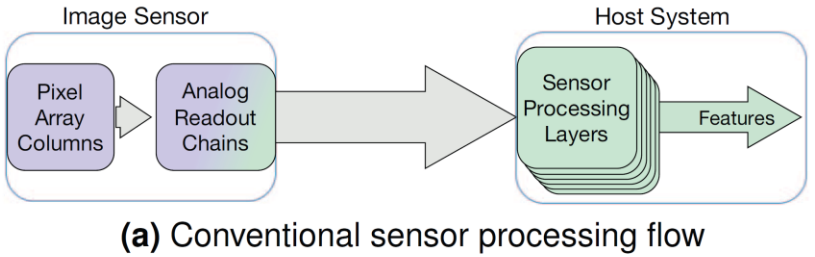




Chen, H. G., Jayasuriya, S., Yang, J., Stephen, J., Sivaramakrishnan, S., Veeraraghavan, A., & Molnar, A. (2016). ASP vision: Optically computing the first layer of convolutional neural networks using angle sensitive pixels. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 903-912).

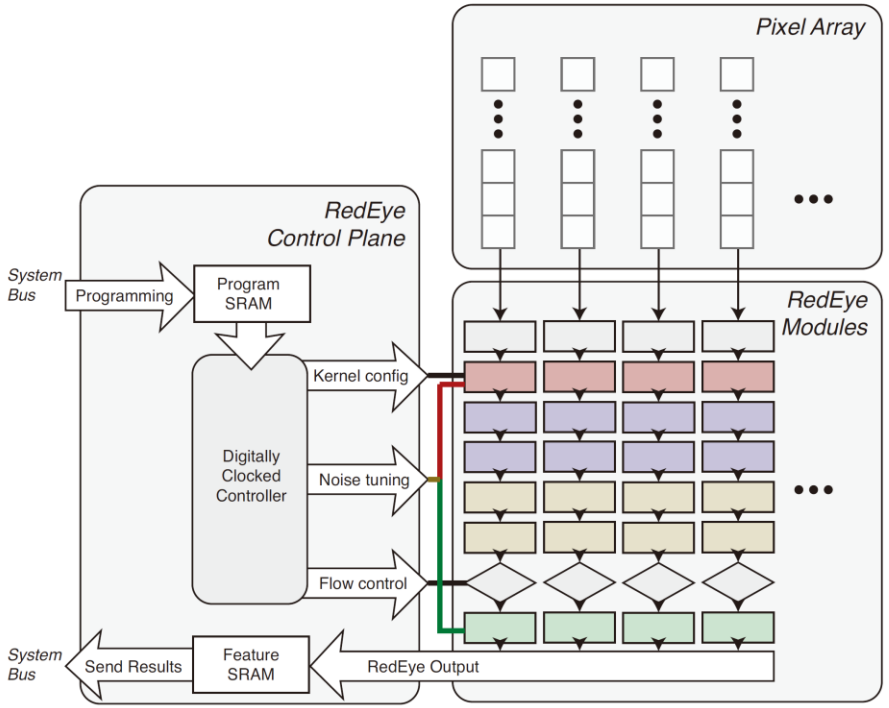
RedEye

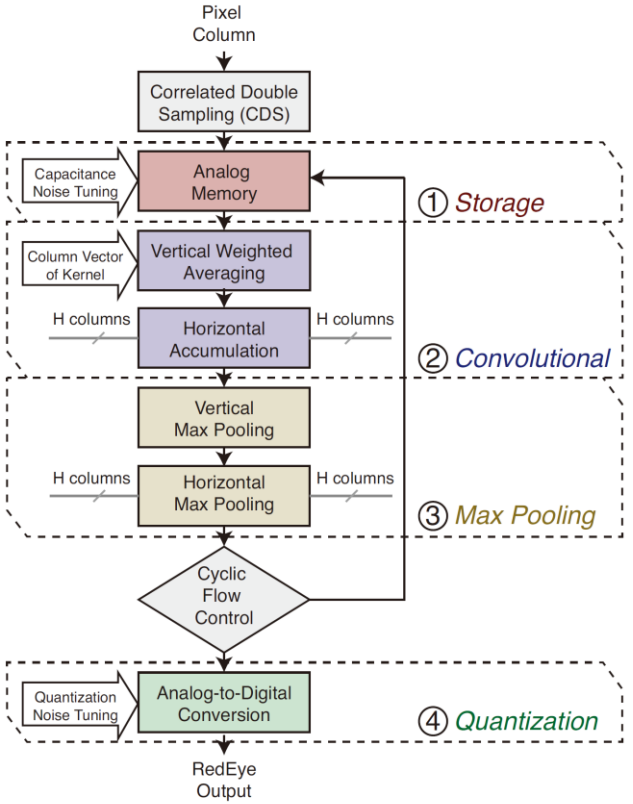
- Analog ConvNet Image Sensor Architecture for Continuous Mobile Vision



(a) Conventional sensor processing flow

(b) RedEye sensor processing flow





LiKamWa, R., Hou, Y., Gao, J., Polansky, M., & Zhong, L. (2016). Redeye: analog convnet image sensor architecture for continuous mobile vision. *ACM SIGARCH Computer Architecture News*, *44*(3), 255-266.

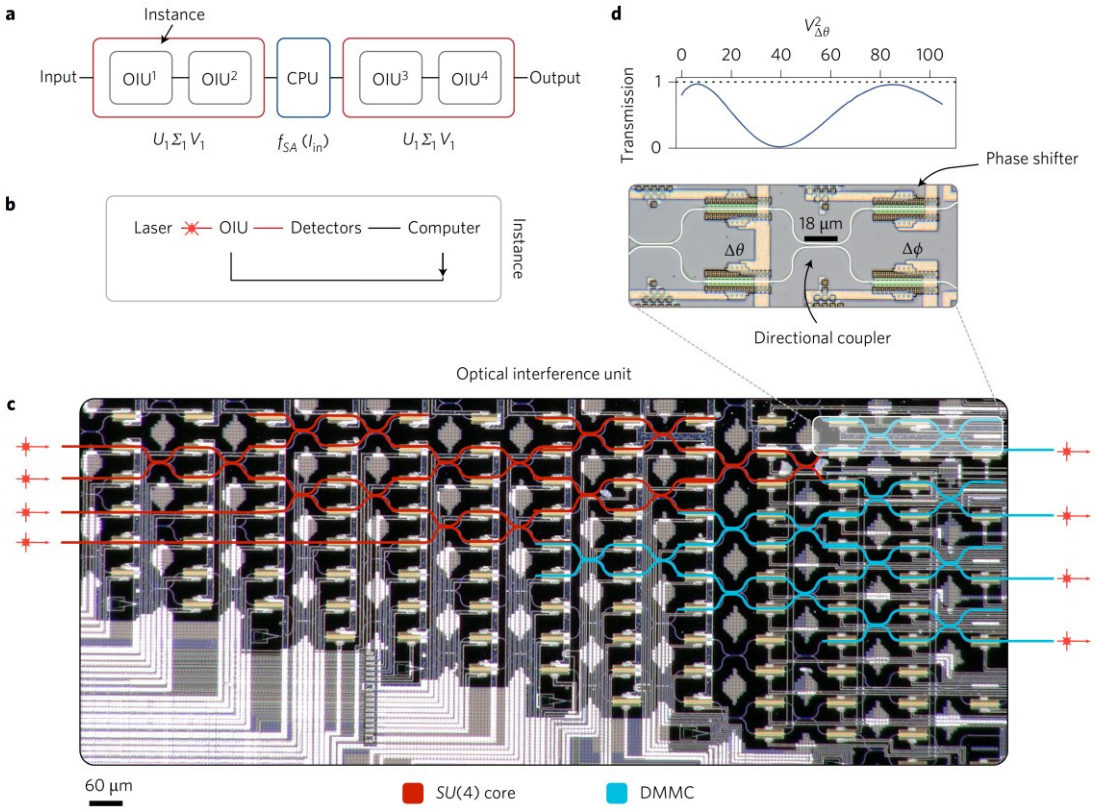
Outline

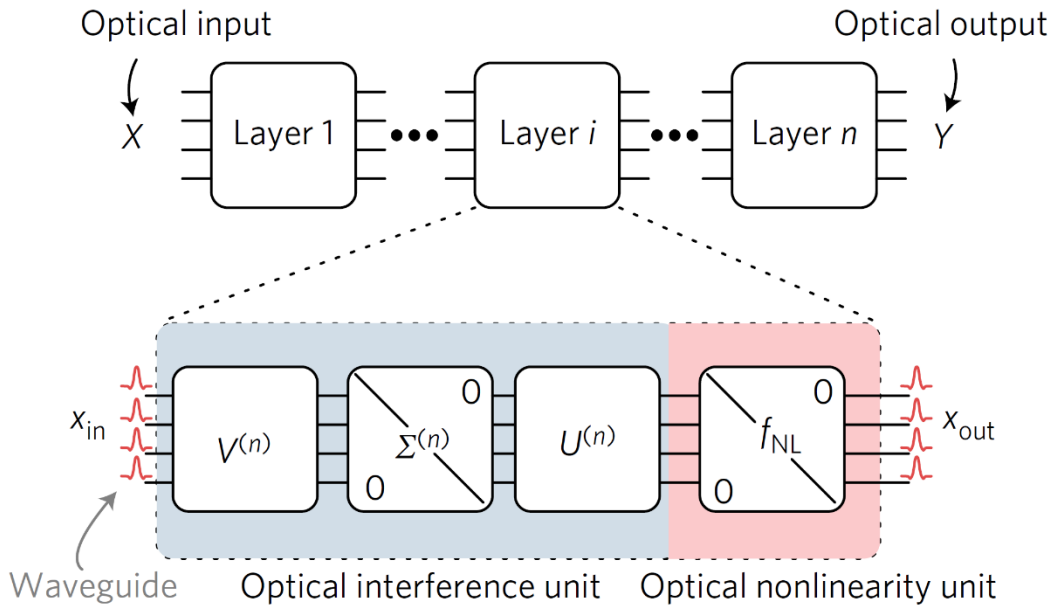
- Processing Near Memory
- Processing in Memory
- Processing in Sensor
- Processing in the Optical Domain
- Accelerator Level Parallelism

Optical Neural Networks

- Matrix Multiplication in the Optical Domain
- The photodetection rate is 100 GHz

*"In principle, such a system can be at least **two orders of magnitude faster** than electronic neural networks (which are restricted to a GHz clock rate)"*

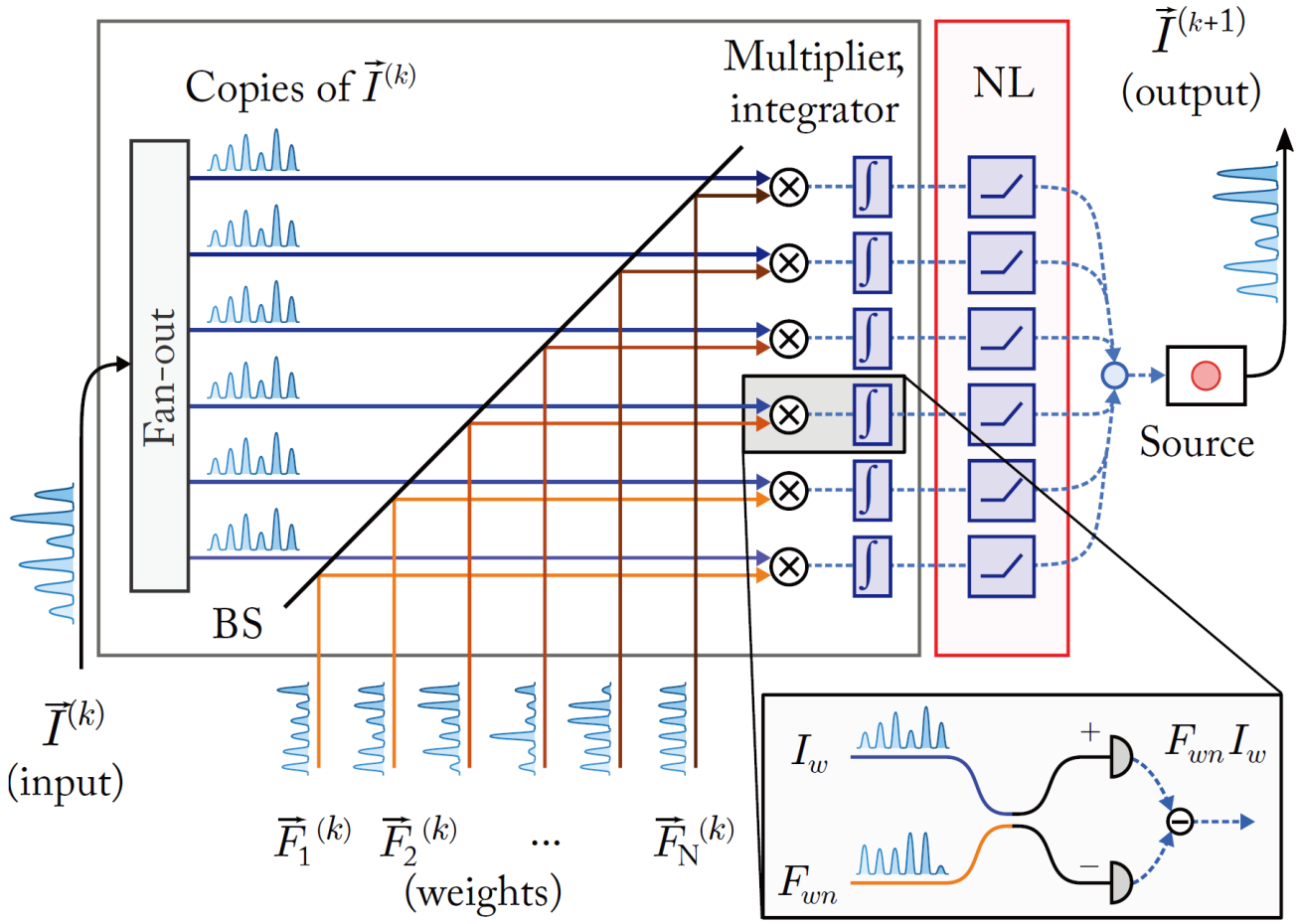




Shen, Y., Harris, N. C., Skirlo, S., Prabhu, M., Baehr-Jones, T., Hochberg, M., ... & Soljačić, M. (2017). Deep learning with coherent nanophotonic circuits. *Nature Photonics*, *11*(7), 441-446.

Optical Neural Network Using Coherent Detection

- BS: beamsplitter
- NL: nonlinearity

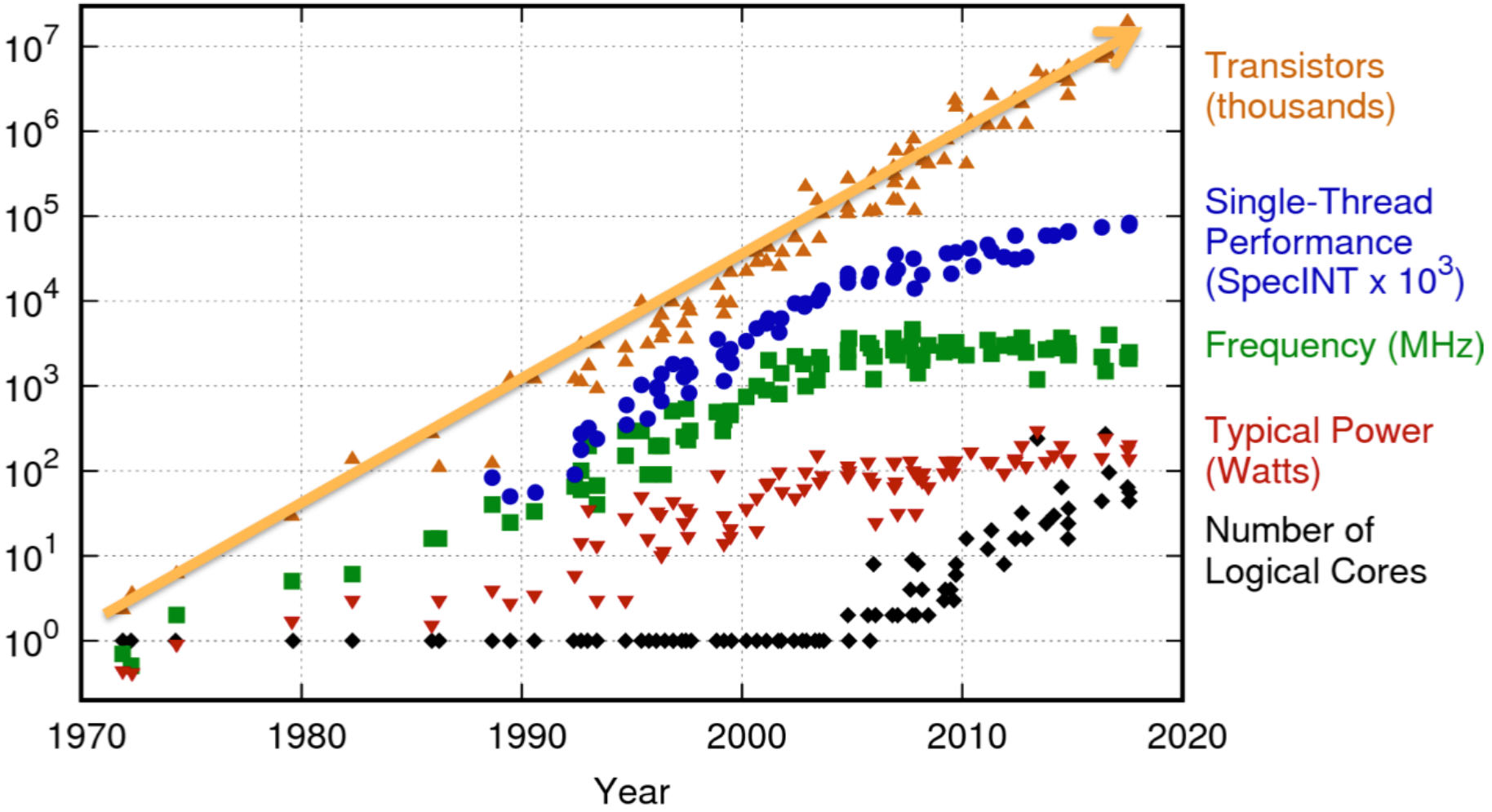


Hamerly, R., Bernstein, L., Sludds, A., Soljačić, M., & Englund, D. (2019). Large-scale optical neural networks based on photoelectric multiplication. *Physical Review X*, *9*(2), 021032.

Outline

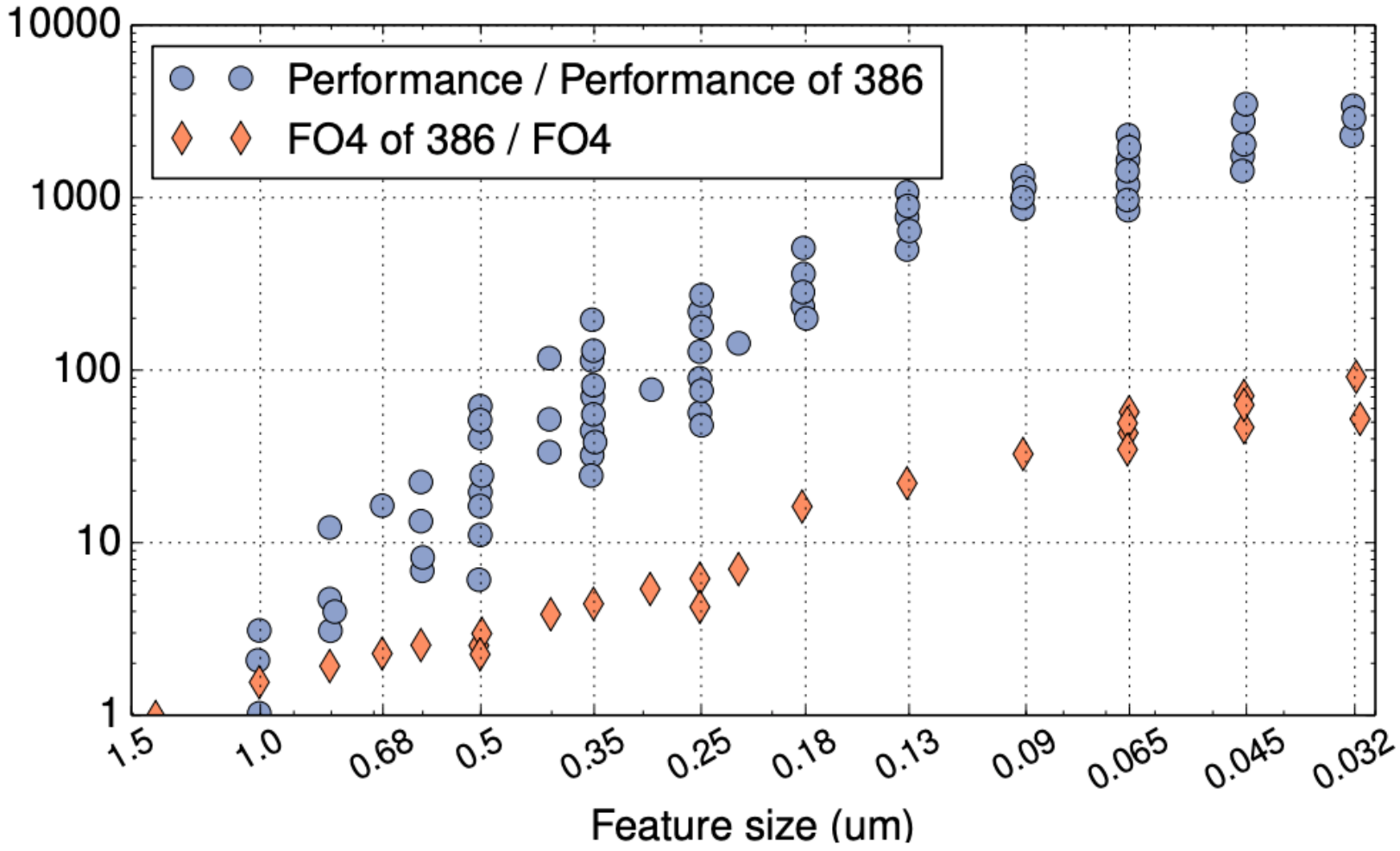
- Processing Near Memory
- Processing in Memory
- Processing in Sensor
- Processing in the Optical Domain
- Accelerator Level Parallelism

Advances in Processor Performance



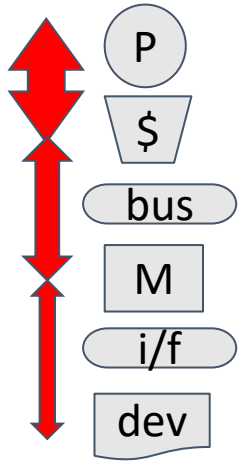
Original data up to the year 2010 collected and plotted by M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, and C. Batten. New plot and data for 2010-2017 by K. Rupp from https://www.karlrupp.net/wp-content/uploads/2018/02/42-years-processor-trend.png

Enablers: Technology and Architecture



Danowitz, A., Kelley, K., Mao, J., Stevenson, J. P., & Horowitz, M. (2012). CPU DB: recording microprocessor history. *Communications of the ACM*, *55*(4), 55-63.

X-Level Parallelism in Computer Architecture



1 CPU

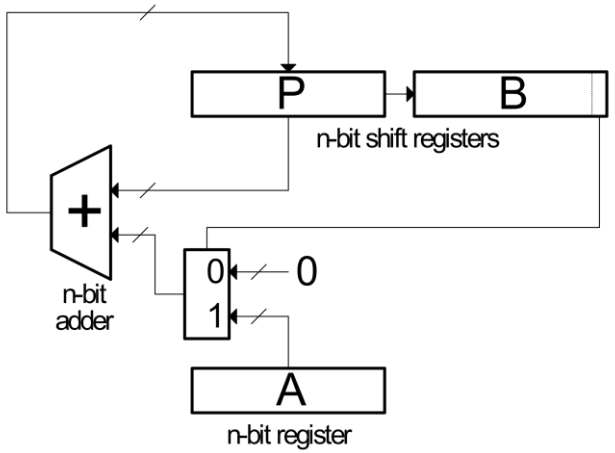
BLP+ILP

Bit/Instrn-Level Parallelism

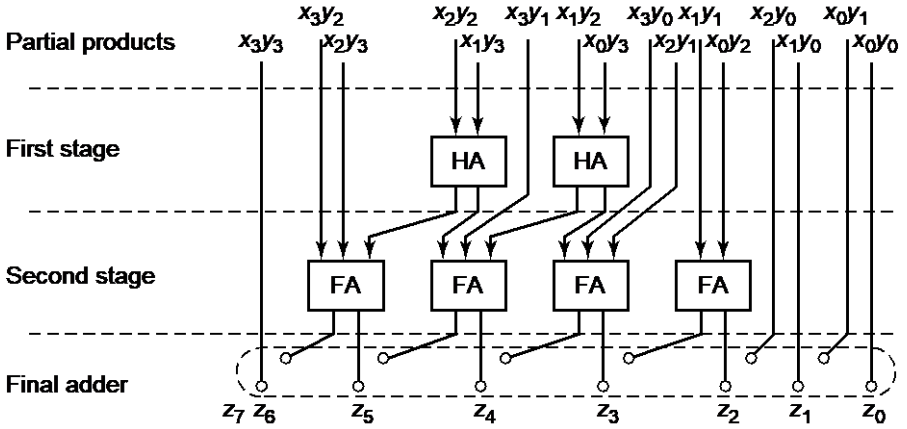
Hill, M. D., & Reddi, V. J. (2021). Accelerator-level parallelism. *Communications of the ACM*, *64*(12), 36-38.

Bit-Level Parallelism (BLP)

- Early computers with few transistors
  - Compute a result in many steps
  - E.g., 1 multiplication partial sum per cycle for “shift-and-add” multiplier
- Bit-Level Parallelism
  - More bits in parallel
  - Larger words improve BLP:
    - 8b->16b->32b->64b
  - Recent trends:
    - Smaller word size preferred for better efficiency, e.g., DNN inference HW.



- “Shift-and-add” multiplier
- Performance: N cycles for N-bit additions



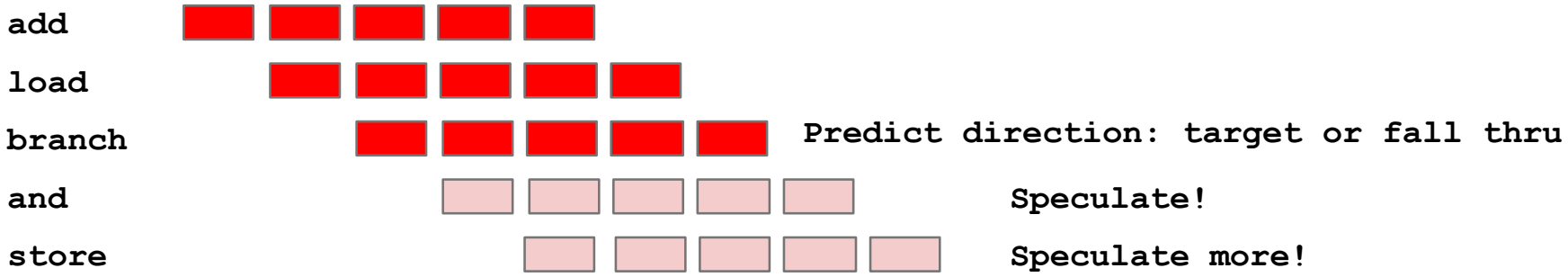
- Parallel Wallace Tree Multiplier

Instruction-Level Parallelism (ILP)

- Processors logically do instructions sequentially (time →)

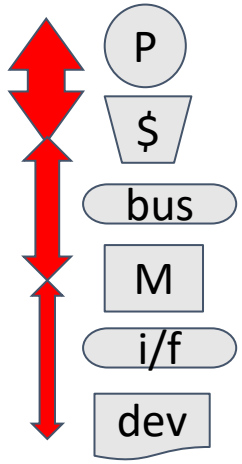


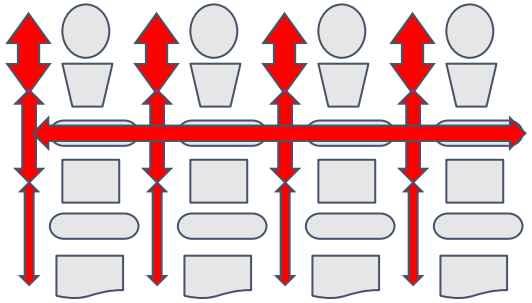
- Though they actually execute instructions in parallel, i.e., instruction-level parallelism.



- Completely hides hardware complexity from software.

X-Level Parallelism in Computer Architecture





1 CPU

BLP+ILP

Bit/Instrn-Level Parallelism

Multiprocessor

+ TLP

Thread-Level Parallelism

Thread-Level Parallelism (TLP)

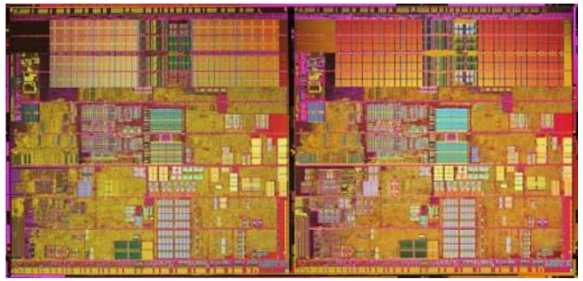
- HW: Multiple sequential processors
- SW: Each runs asynchronous thread

- SW must partition work, synchronize, & manage communication
  - E.g., pThreads, OpenMP, MPI

- Less easy for software but:
  - Bifurcation: experts program TLP; others use it.

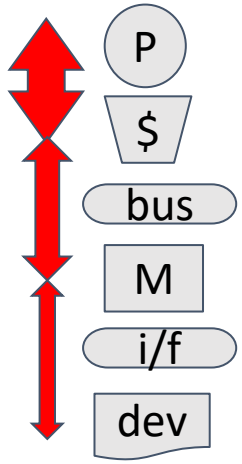


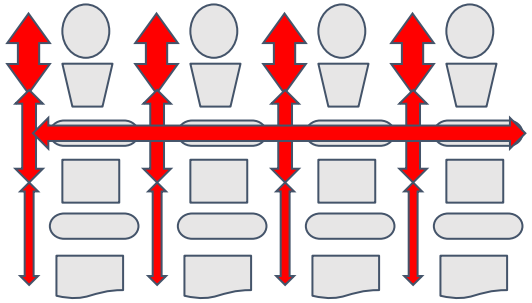
CDC 6600, 1964,
(TLP via multithreaded processor)

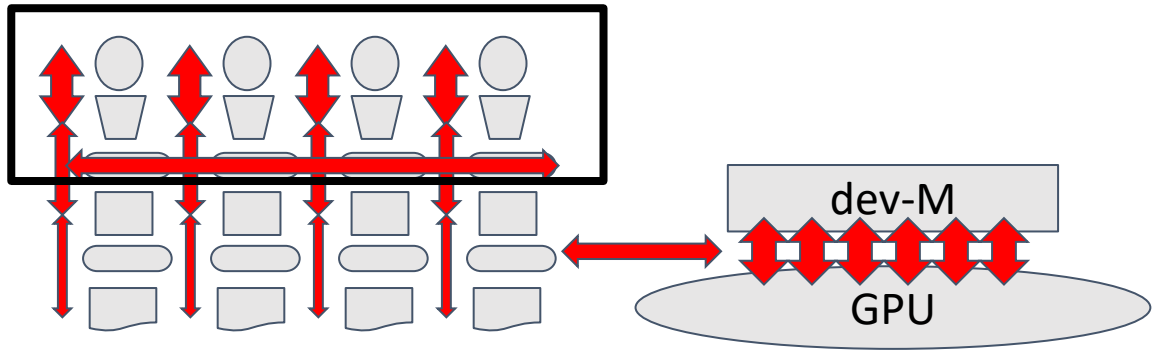


Intel Pentium Pro Extreme Edition,
early 2000s

X-Level Parallelism in Computer Architecture

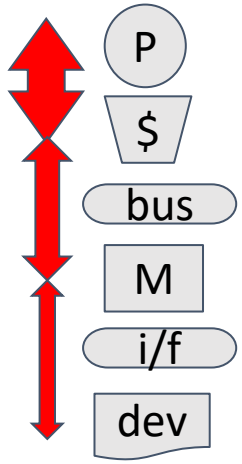


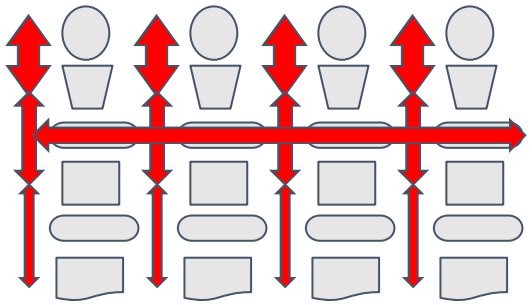


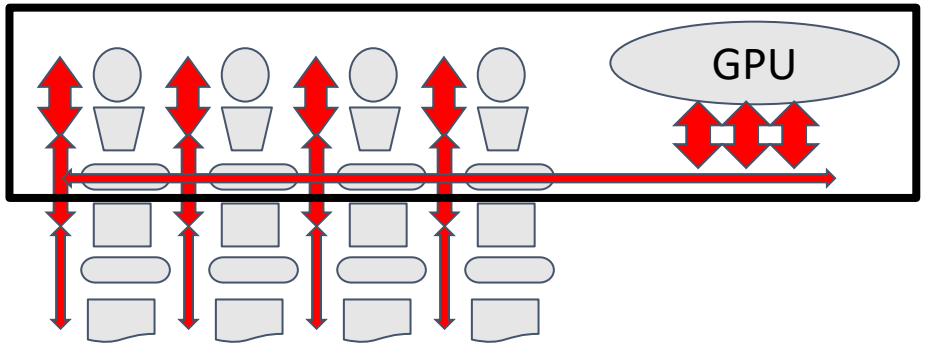


1 CPU	Multiprocessor	Multicore	+ Discrete GPU
BLP+ILP	+ TLP	+ TLP	+ DLP
Bit/Instrn-Level Parallelism	Thread-Level Parallelism	Thread-Level Parallelism	Data-Level Parallelism

X-Level Parallelism in Computer Architecture







1 CPU

BLP+ILP

Bit/Instrn-Level Parallelism

Multiprocessor

+ TLP

Thread-Level Parallelism

Multicore + **Integrated GPU**

+ TLP

Thread-Level Parallelism

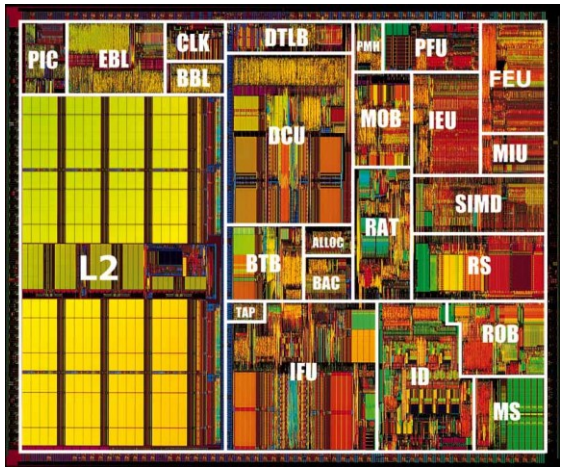
+ DLP

Data-Level Parallelism

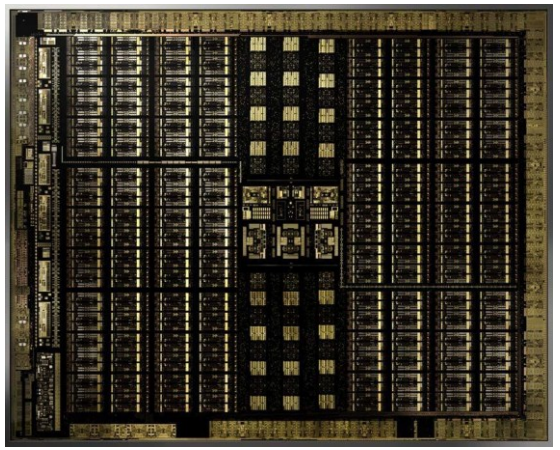
Data-Level Parallelism

- Perform the same operation on different data elements
  - Single-Instruction-Multiple-Data (SIMD)

- DLP success: General-Purpose GPUs
  - Single-Instruction-Multiple-Threads (SIMT)
  - SW (CUDA) & Libraries (math & ML)
  - Bifurcation again:
    - Experts program SIMT (TLP + DLP);
    - Others use it.

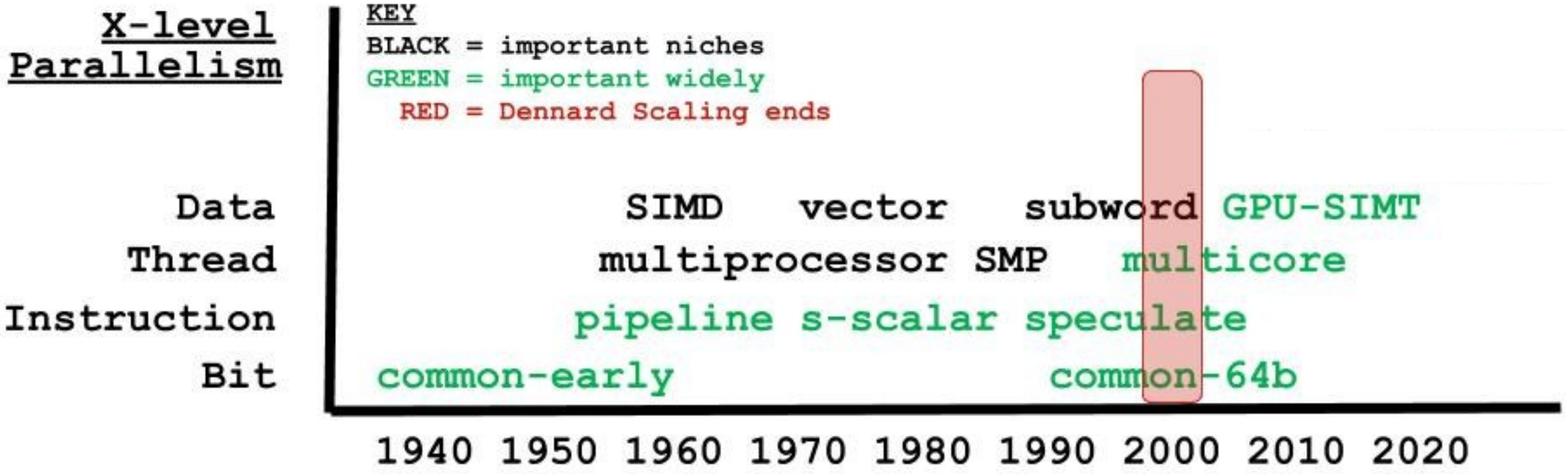


Intel Pentium III



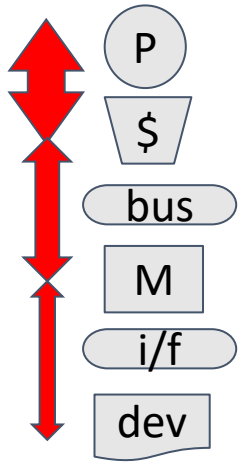
NVIDIA Turing

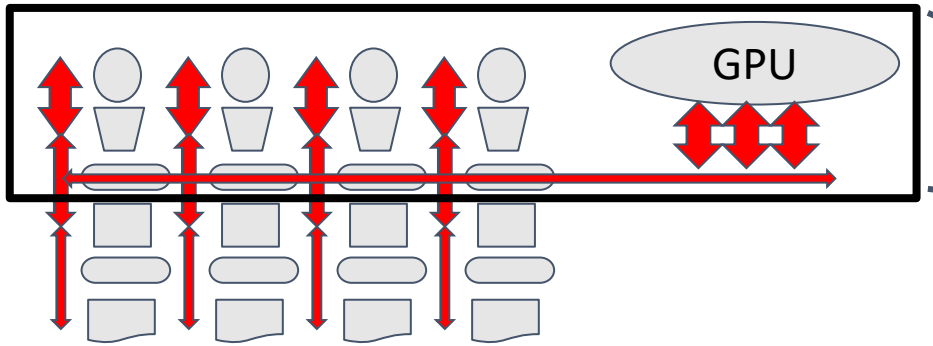
X-Level Parallelism in Computer Architecture

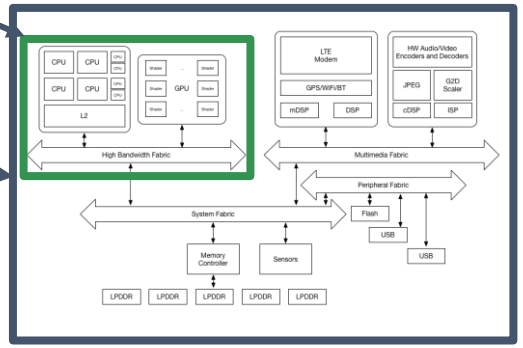


X-Level Parallelism in Computer Architecture

- Accelerator-level parallelism (ALP): the parallelism of different workload components concurrently executing on multiple accelerators.







1 CPU

BLP+ILP

Bit/Instrn-Level Parallelism

Multicore + **Integrated GPU**

+ TLP

Thread-Level Parallelism

+ DLP

Data-Level Parallelism

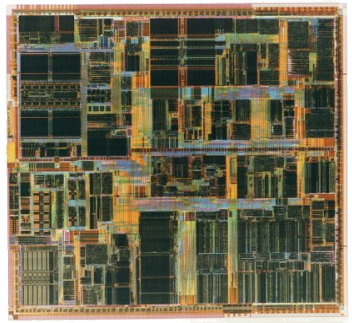
System on a Chip (SoC)

+ ALP

Accelerator-Level Parallelism

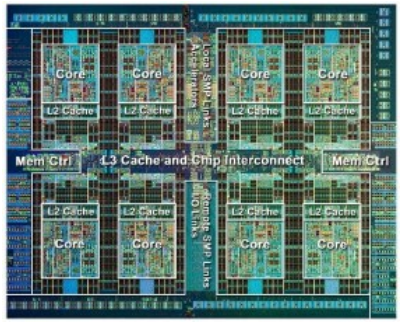
X-level Parallel Hardware + Software

Single-threaded languages, compilers, runtimes, etc. **hides** parallelism



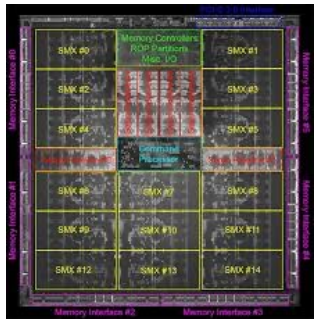
Intel Pentium Pro
BLP+ILP

Software **abstracts** TLP, e.g., pThreads, OpenMP, & MPI. Also **hidden** in cloud, etc.



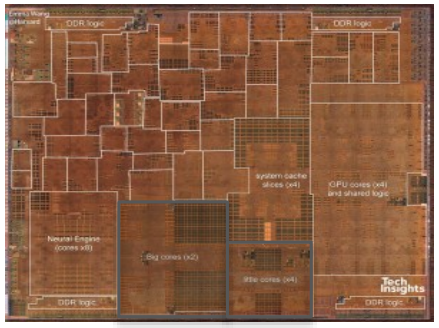
IBM Power 7
BLP+ILP+TLP

Software **abstracts** TLP+DLP, e.g., CUDA, OpenCL, & graphics OpenGL Also **hidden** in cloud, etc.



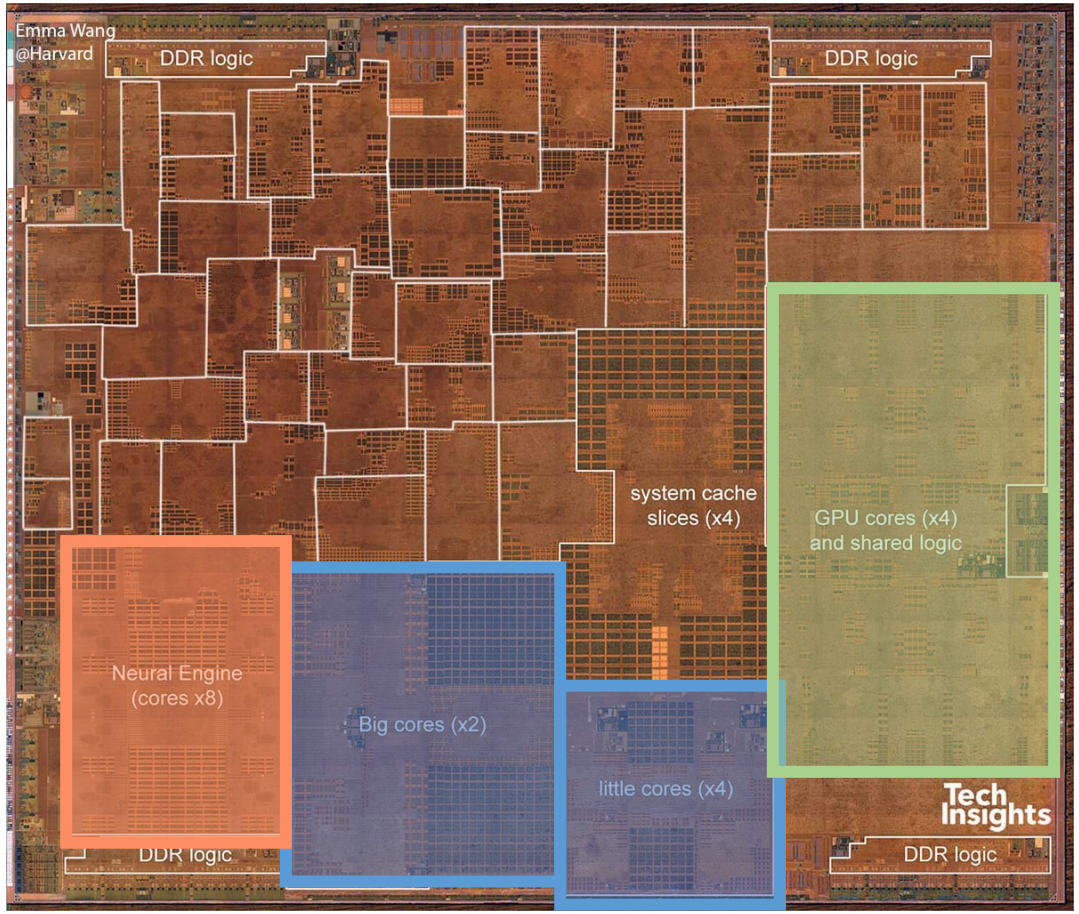
Nvidia GK110
BLP+TLP+DLP

Local software stack **abstracts** each accelerator.
But no good, general software abstraction for SoC ALP!

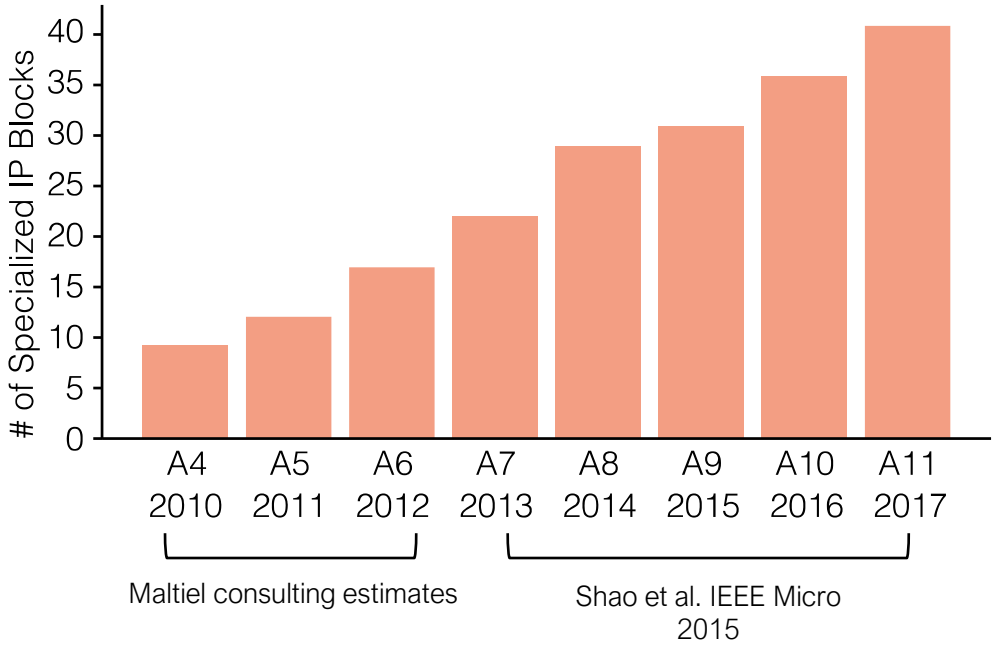


Apple A12
BLP+ILP+TLP+DLP+ALP

CPU, GPU, and Accelerators



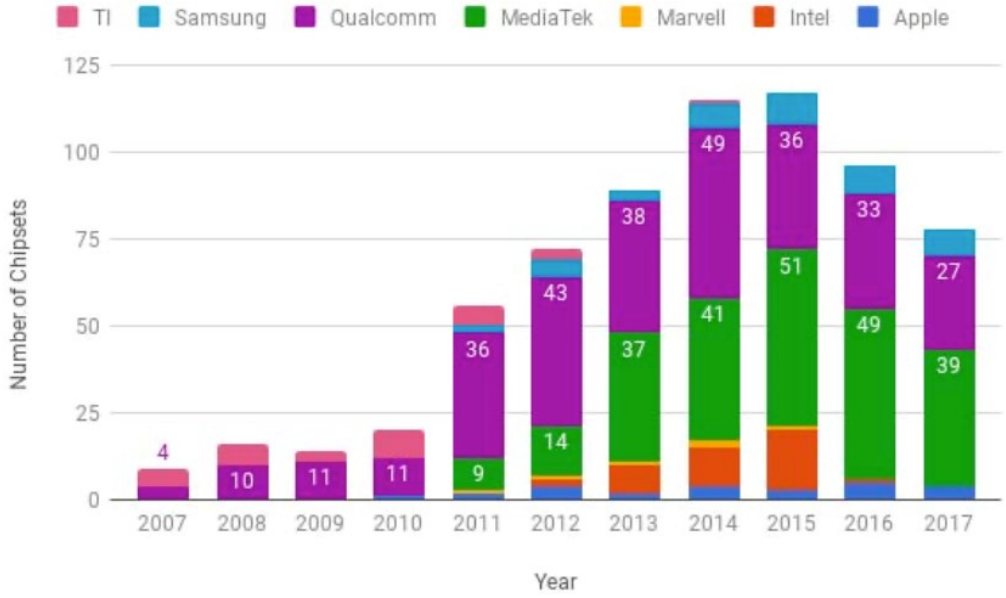
Apple A12 SoC



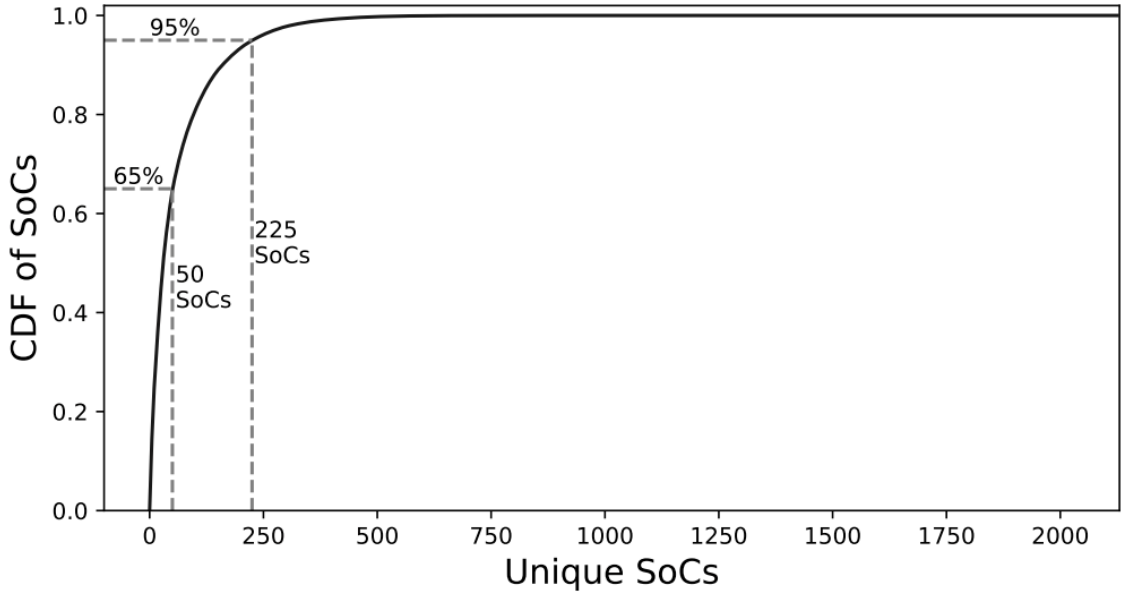
http://vlsiarch.eecs.harvard.edu/research/accelerators/die-photo-analysis/

Mobile SoC Diversity

- 10s-100s of SoCs introduced to the market every years across different SoC vendors.
- No standard mobile SoC to optimize for. Top 50 most common SoCs -> only 65% of the market.



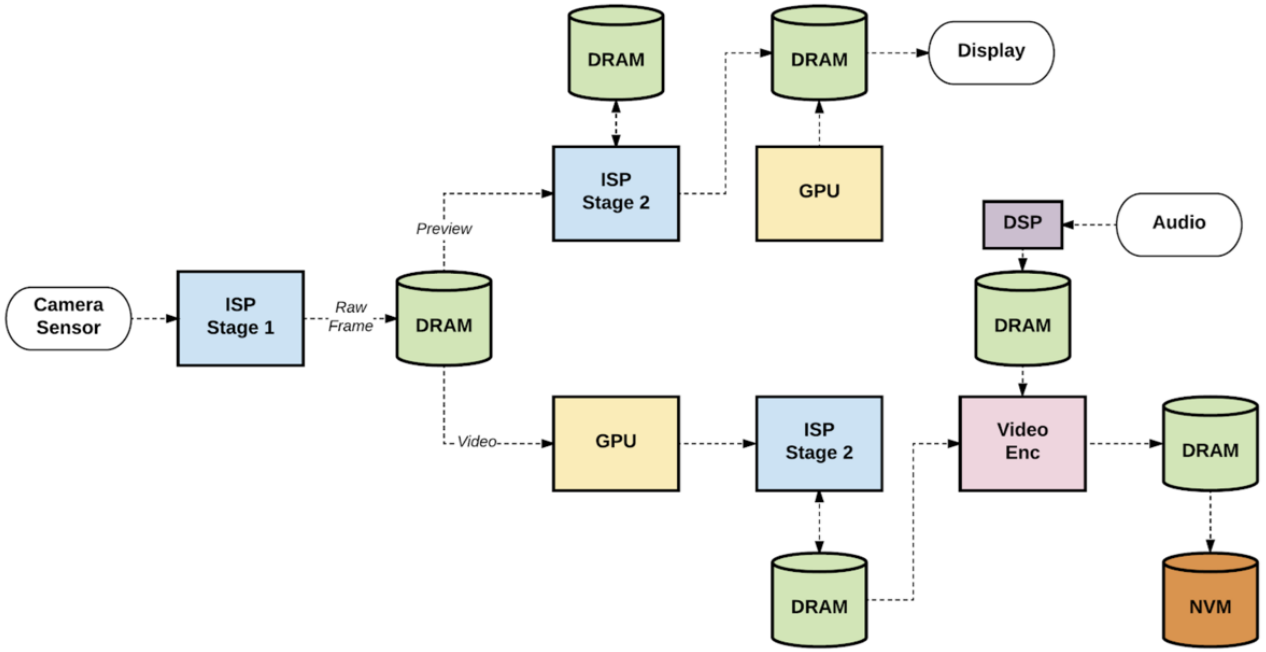
Two Billion Devices and Counting: An Industry Perspective on the State of Mobile Computer Architecture, IEEE Micro'2018



Machine Learning at Facebook: Understanding Inference at the Edge, HPCA'2019

Mobile SoC Usecases

- Mainstream architecture has long focused on general-purpose CPUs and GPUs.
- In an SoC, multiple IP blocks are active at the same time and communicate frequently with each other.
- Example:
  - Recording a 4K video
  - Camera -> ISP
    - “Preview stream” for display
    - “Video stream” for storage
  - DRAM for data sharing



Two Billion Devices and Counting: An Industry Perspective on the State of Mobile Computer Architecture, IEEE Micro’2018

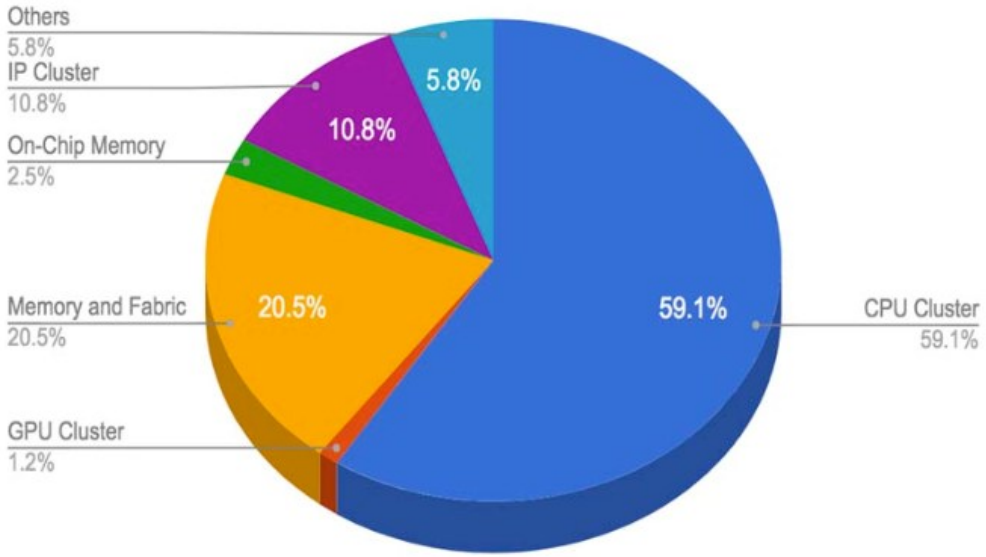
Mobile SoC Usecases

- Multiple accelerators are running concurrently for different usecases.

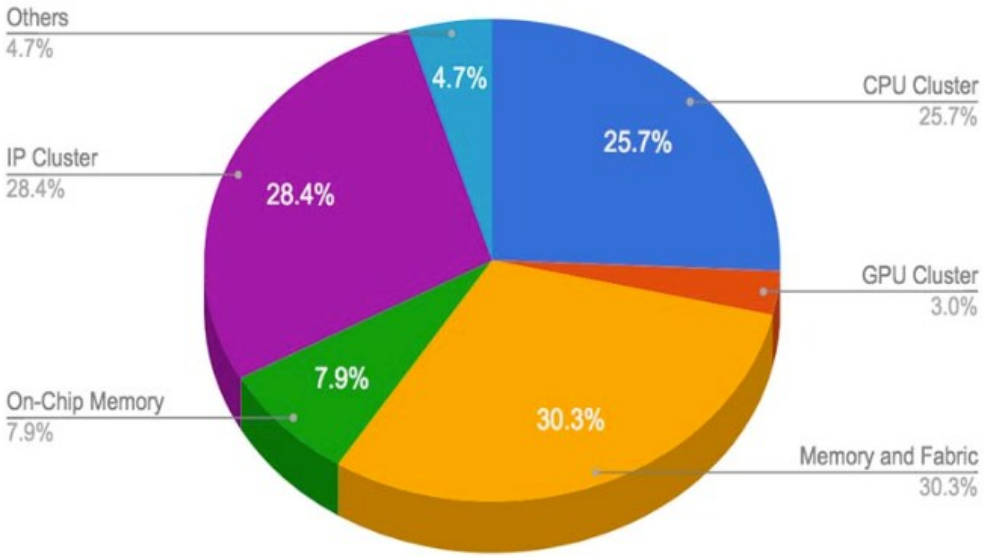
Accelerators (IPs) ➔ Usecases (rows)	CPUs (AP)	Display	Media Scaler	GPU	Image Signal Proc.	JPEG	Pixel Visual Core	Video Decoder	Video Encoder	**Dozens More**
Photo Enhancing	X	X		X	X	X	X			
Video Capture	X	X		X	X				X	
Video Capture HDR	X	X		X	X				X	
Video Playback	X	X	X	X				X		
Image Recognition	X	X	X	X						

Mobile SoC Usecases

- Different usecases have different power profiles.



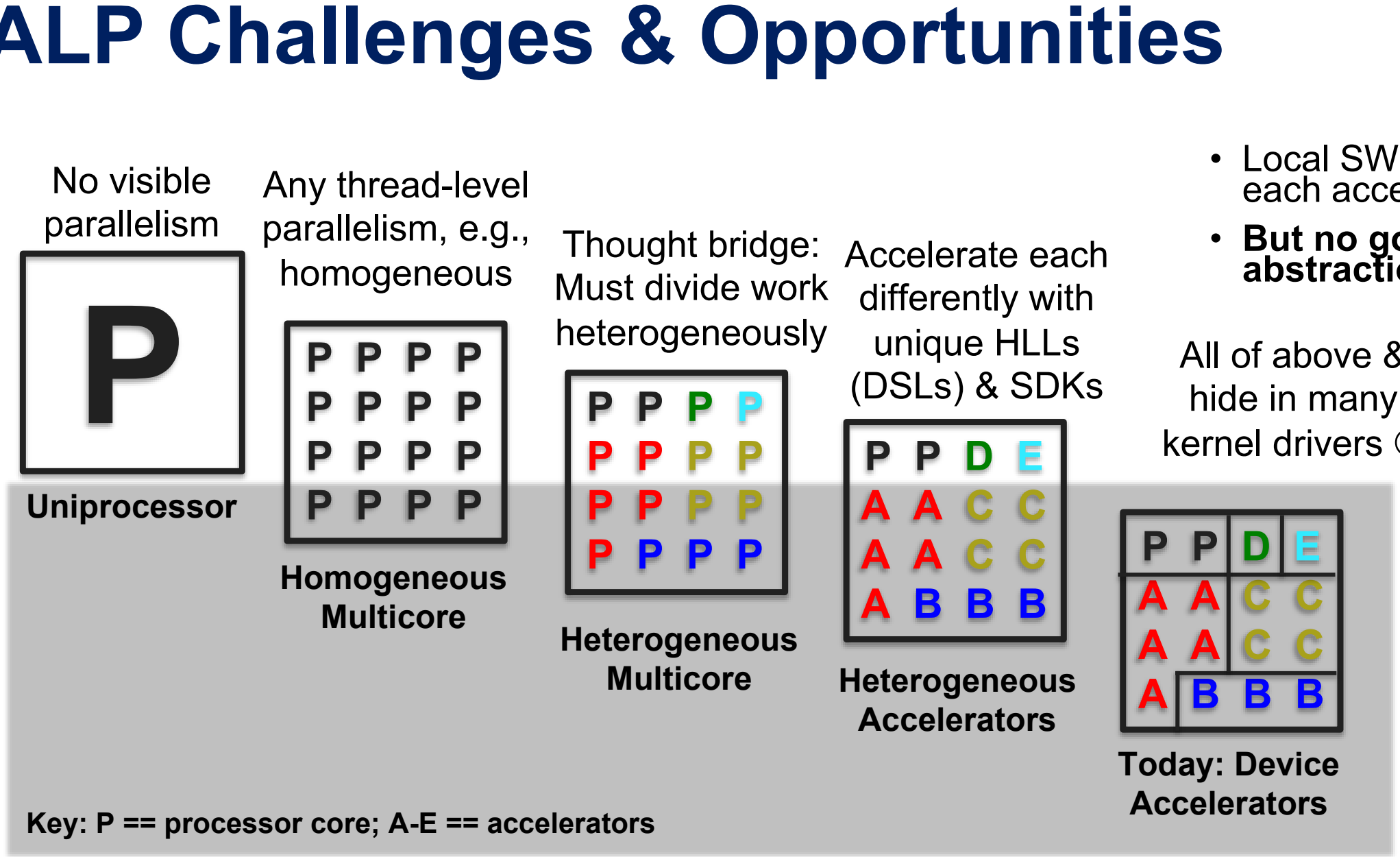
Web browser scrolling
(total 651.77 mW)



4K video recording
(total 3,473.55mW)

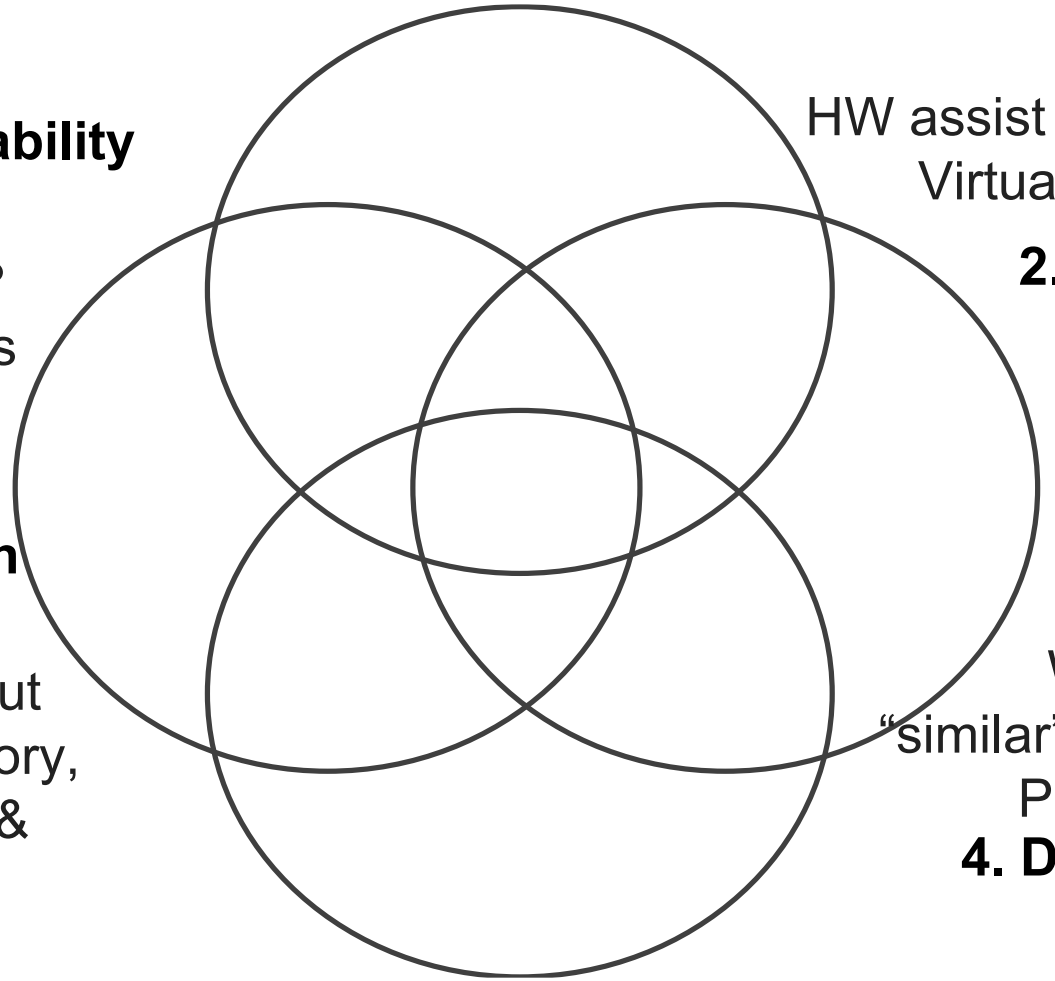
Reddi, V. J., Yoon, H., & Knies, A. (2018). 2 Billion Devices and Counting: An Industry Perspective on the State of Mobile Computer Architecture.

ALP Challenges & Opportunities



- Local SW stack **abstracts** each accelerator.
- **But no good, general SW abstraction for SoC ALP!**

ALP Challenges & Opportunities



1. **Programmability**

Whither global model/runtime? DAG of streams for SoCs?

2. **Concurrency**

HW assist for scheduling? Virtualize & partition?

3. **Communication**

How should SW stack reason about local/global memory, caches, queues, & scratchpads?

4. **Design Space**

When combine "similar" accelerators? Power vs. area?